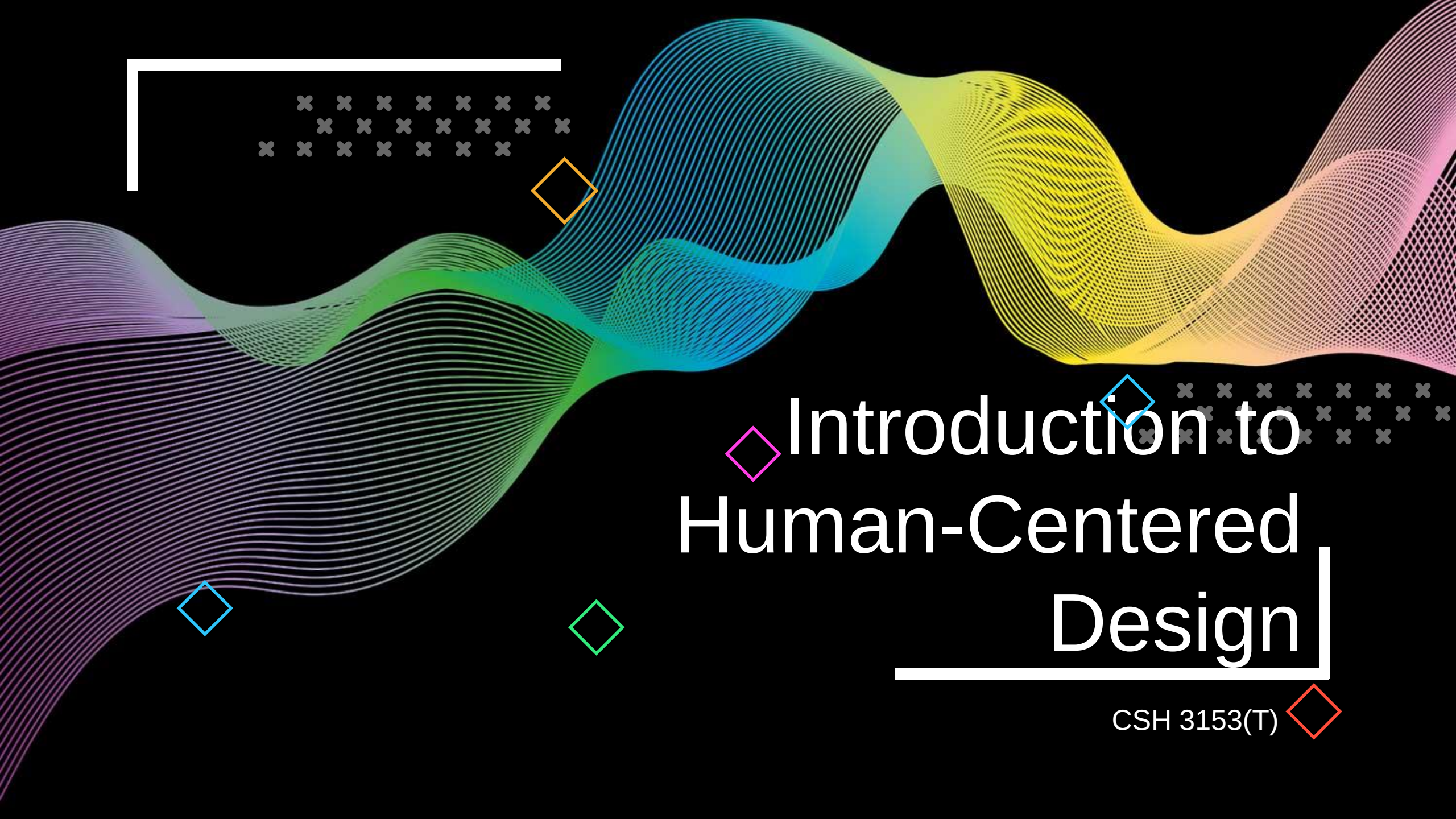


The background features a series of overlapping, wavy lines in shades of purple, green, blue, yellow, and pink. Scattered throughout are small grey 'x' marks and several outlined diamond shapes in orange, pink, blue, green, and red. A white L-shaped line is positioned in the top-left corner, and another white L-shaped line is at the bottom-right, framing the title and course code.

Introduction to Human-Centered Design

CSH 3153(T)

definition

Human-Centered Design (HCD) is an approach to designing products, systems, and services that focuses on understanding the needs, preferences, and behaviors of the end users. It places the human perspective at the core of the design process.



Objectives

HCD aims to create solutions that are not just functional but also user-friendly, efficient, and satisfying.

It seeks to minimize the gap between user expectations and system behavior.



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Historical Evolution of HCD

Origins in Industrial Design and Ergonomics

Human-Centered Design has its roots in industrial design and ergonomics, which focused on creating products and environments that are comfortable and efficient for human use.

Pioneers like Henry Dreyfuss and Raymond Loewy contributed to designing products with a strong emphasis on user experience and functionality.

Concepts such as anthropometrics (study of human body measurements) and usability testing began to shape the field.



Historical Evolution of HCD

Emergence in Software and Interface Design

HCD gained prominence in the field of software and interface design as computers became more accessible to the general population.

The shift from command-line interfaces to graphical user interfaces (GUIs) in the 1980s marked a significant milestone.

Researchers like Alan Cooper and Don Norman made valuable contributions, emphasizing user-friendly software design and the importance of user-centered approaches.

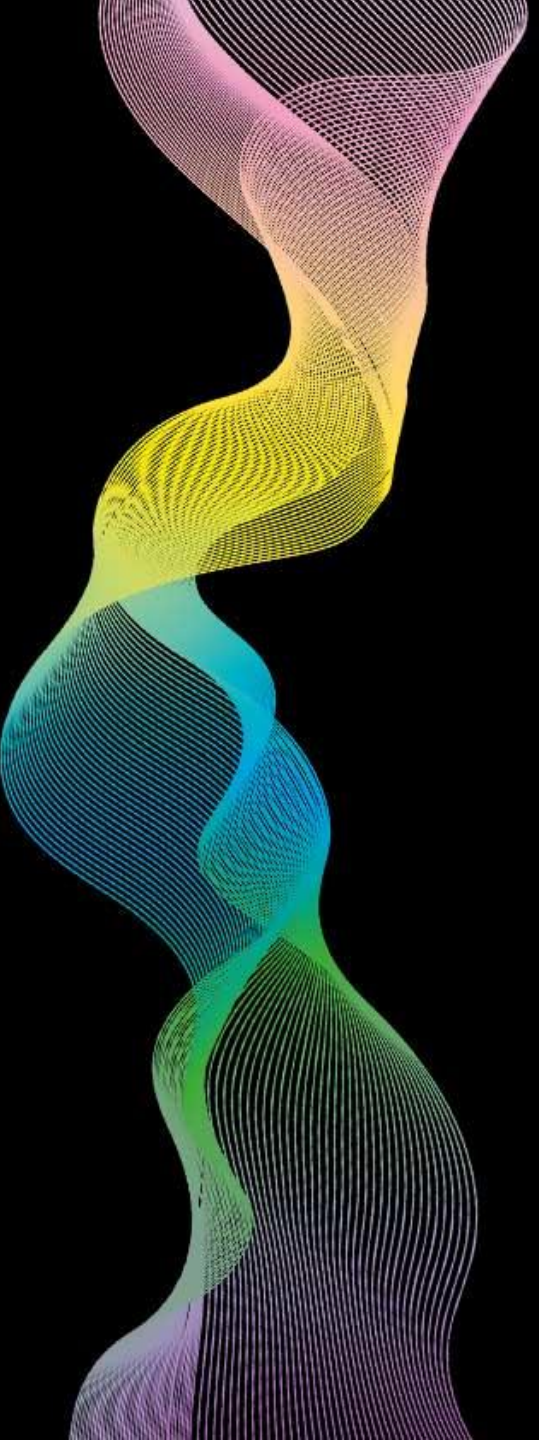


Historical Evolution of HCD

Adoption in Various Domains:

Over time, HCD principles expanded beyond software and interface design into various domains:

- In healthcare, HCD principles are used to design medical devices, patient experiences, and healthcare environments for improved patient outcomes.
- In the automotive industry, HCD is applied to create user-friendly dashboard interfaces, safety features, and vehicle ergonomics.
- In technology, consumer electronics, and product design, HCD is instrumental in designing smartphones, wearable devices, and smart home appliances.
- The adoption of HCD in these diverse domains highlights its adaptability and effectiveness in enhancing user experiences and outcomes.

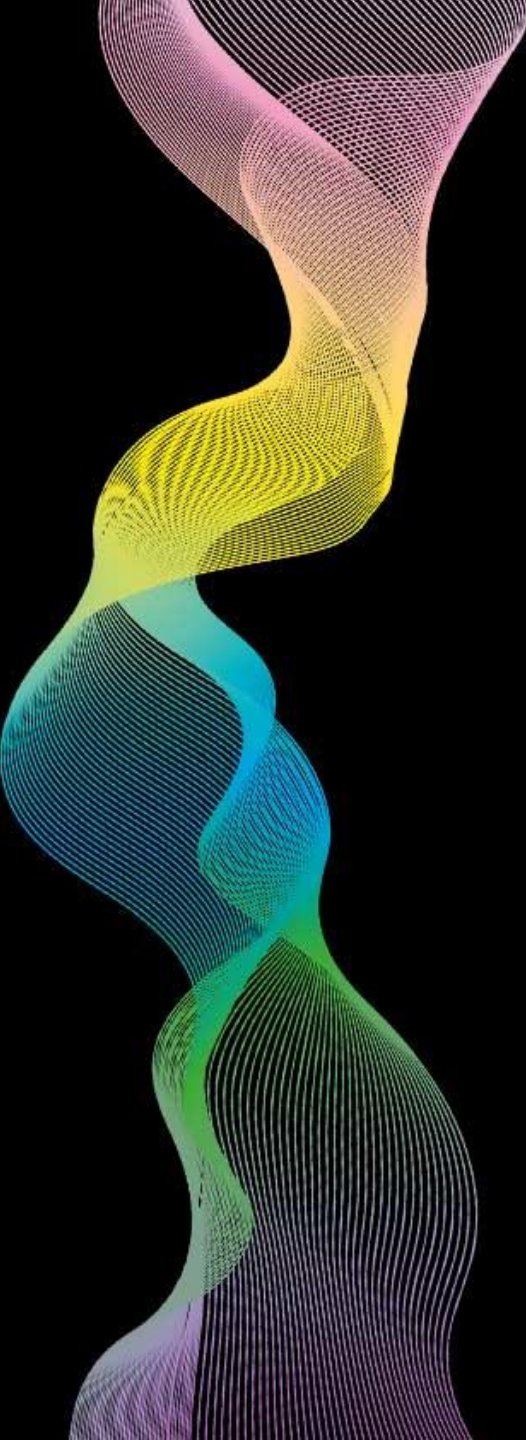


Why Human-Centered Design Matters?

Improved User Satisfaction and Engagement

HCD prioritizes understanding and meeting the needs and preferences of users. By designing with users in mind, products and services become more aligned with what users want and expect.

This alignment leads to higher levels of user satisfaction and engagement. Users are more likely to enjoy using a product or service that feels intuitive, useful, and tailored to their needs.

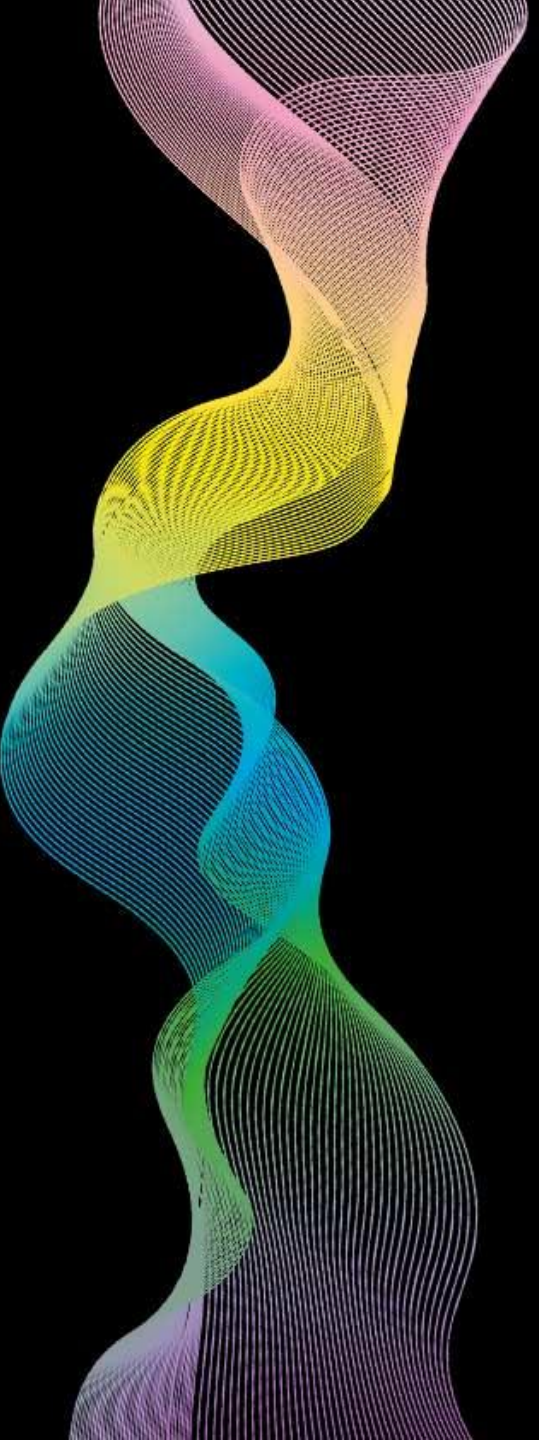


Why Human-Centered Design Matters?

Reduced Errors and User Frustration

When products and services are designed without considering the user's perspective, usability issues, and errors can arise. Users may struggle to complete tasks, leading to frustration and dissatisfaction.

HCD actively identifies and mitigates usability issues through methods like user testing, feedback collection, and iterative design. This results in fewer errors and a smoother user experience.

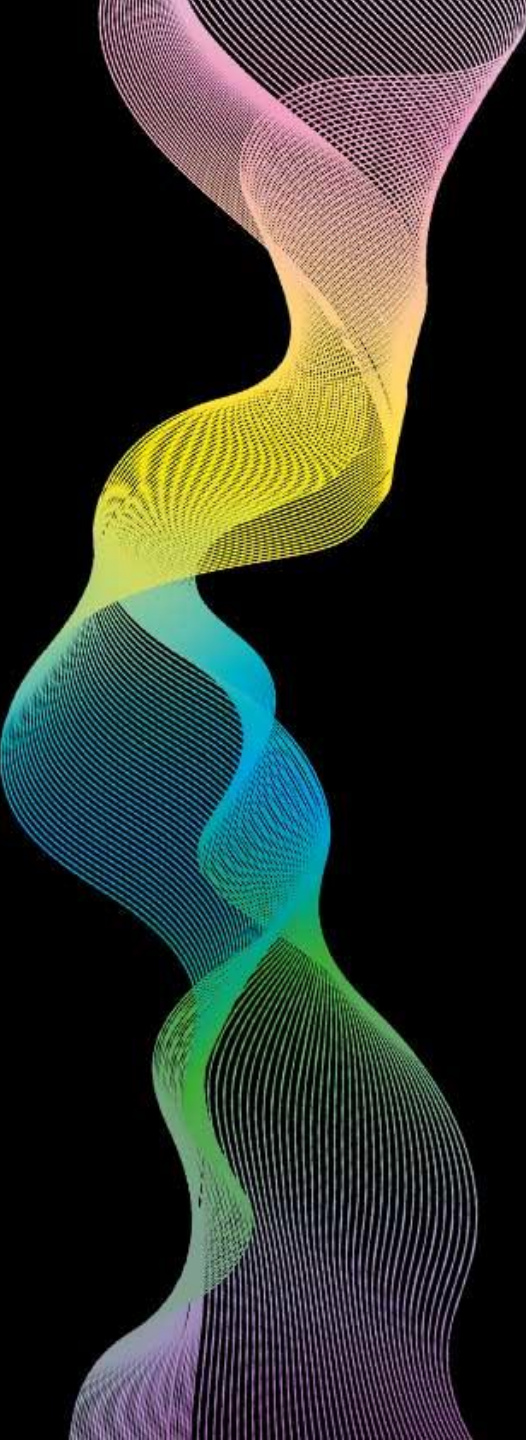


Why Human-Centered Design Matters?

Enhanced Product Adoption and Usability

User adoption is a critical factor in the success of a product or service. If users find a product difficult to use or unintuitive, they may abandon it.

HCD ensures that products are designed with usability in mind, making it easier for users to learn and use them. This, in turn, enhances product adoption rates and reduces barriers to entry.

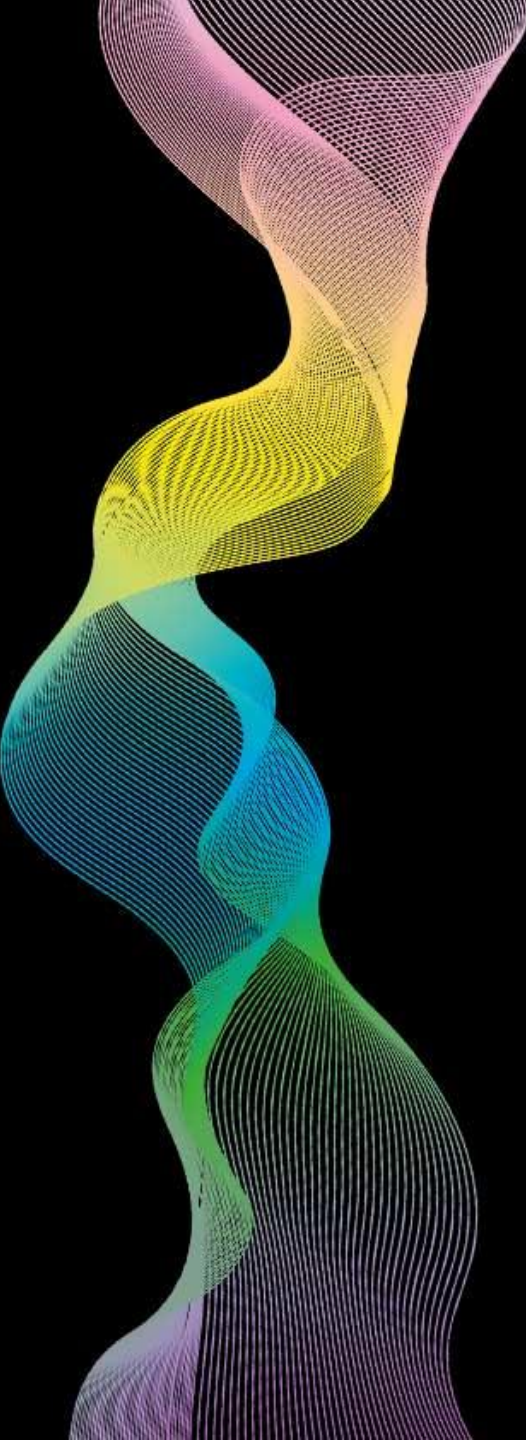


Why Human-Centered Design Matters?

Greater Market Competitiveness

In today's competitive landscape, companies that invest in HCD gain a competitive edge. Products that prioritize user needs and preferences are more likely to stand out and succeed in the market.

HCD-driven products often receive positive reviews and word-of-mouth recommendations, contributing to market competitiveness.

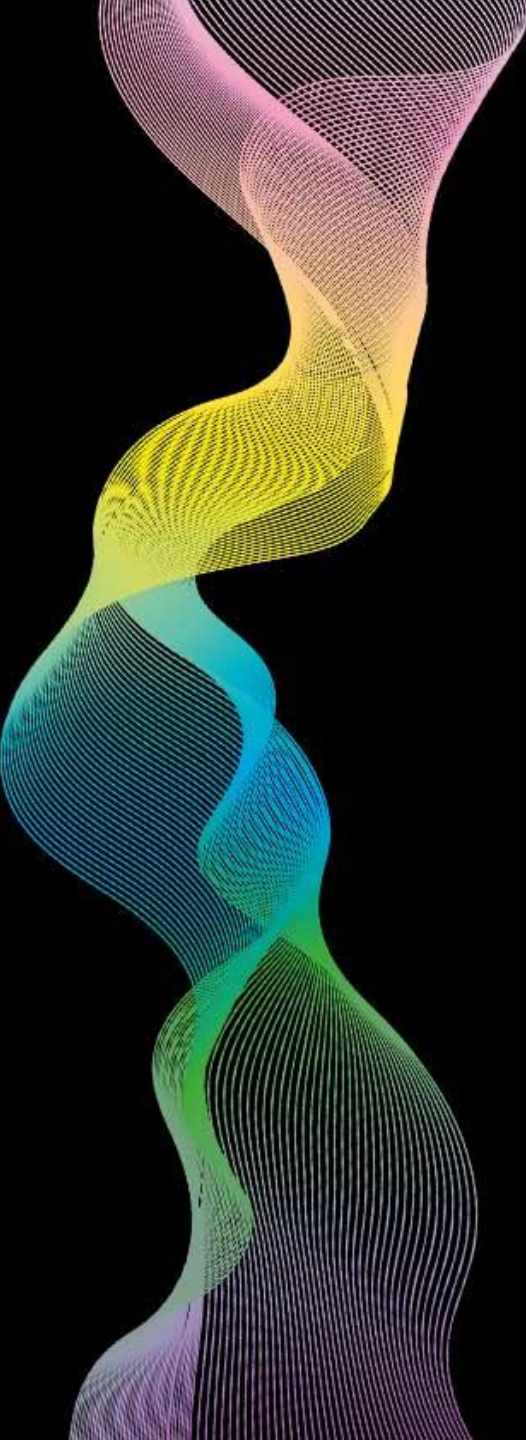


Why Human-Centered Design Matters?

Customer Loyalty and Brand Reputation

HCD fosters positive user experiences, which can lead to increased customer loyalty. Users who have good experiences are more likely to return to the same product or brand.

Positive experiences also contribute to a strong brand reputation, as users associate the brand with quality and user-centricity.



Why Human-Centered Design Matters?

Adaptability to Changing Needs

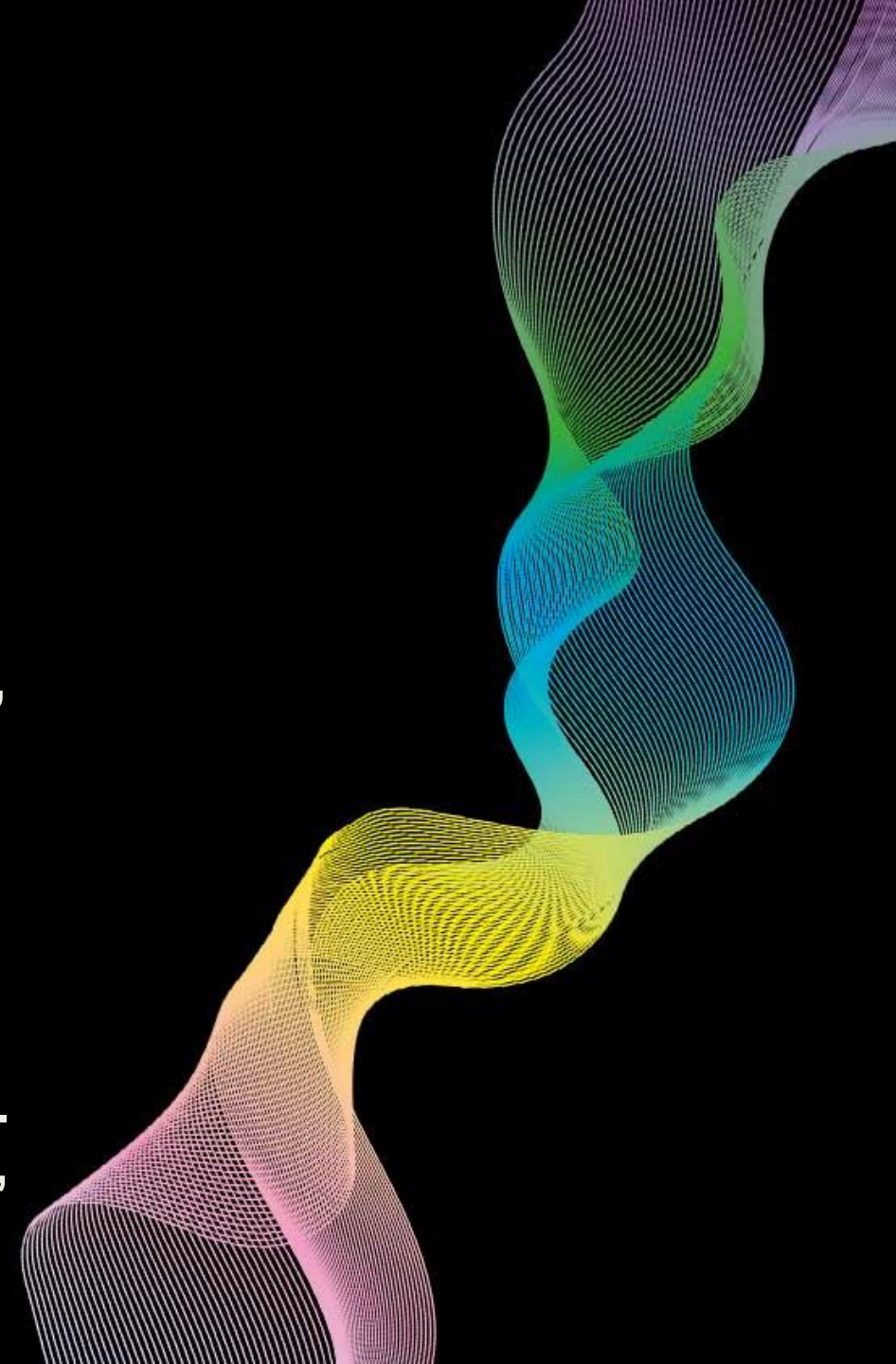
User needs and expectations evolve over time. HCD principles emphasize flexibility and the ability to adapt to changing user requirements.

Products and services designed with HCD can more easily incorporate updates and changes based on user feedback and evolving technology trends.

Examples of Human-Centered Design Successes

Smartphone Interfaces Designed for Intuitive Use

- **Apple iPhone:** The iPhone is a classic example of HCD. Its user interface is known for its simplicity, ease of use, and intuitive design. Features like touch gestures, app organization, and accessibility options have been designed with the user in mind.
- **Android OS:** Android, the mobile operating system developed by Google, also incorporates HCD principles. It offers customization options, user-friendly navigation, and a focus on user preferences.



Examples of Human-Centered Design Successes

Smartphone Interfaces Designed for Intuitive Use

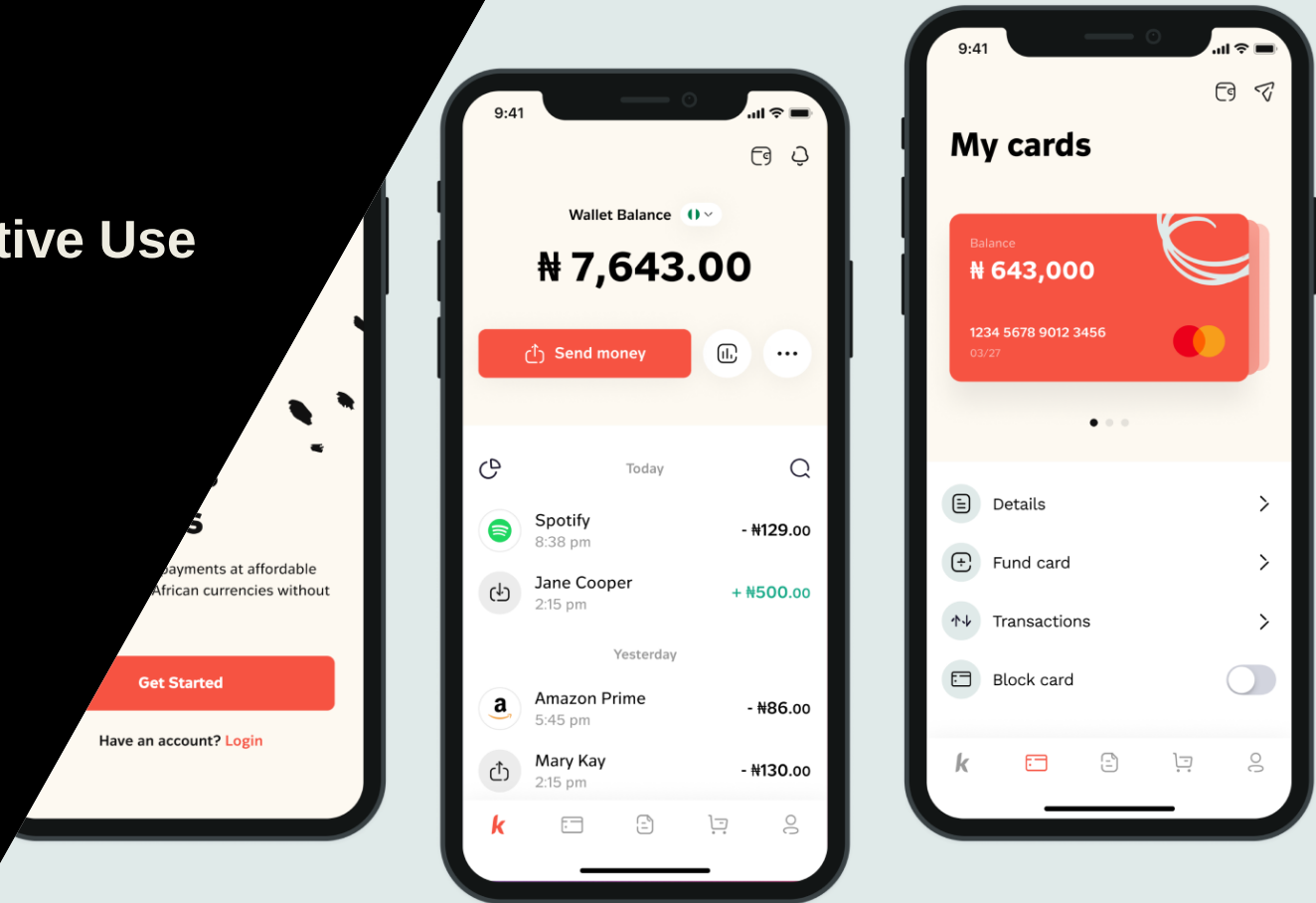
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Examples of Human-Centered Design Successes

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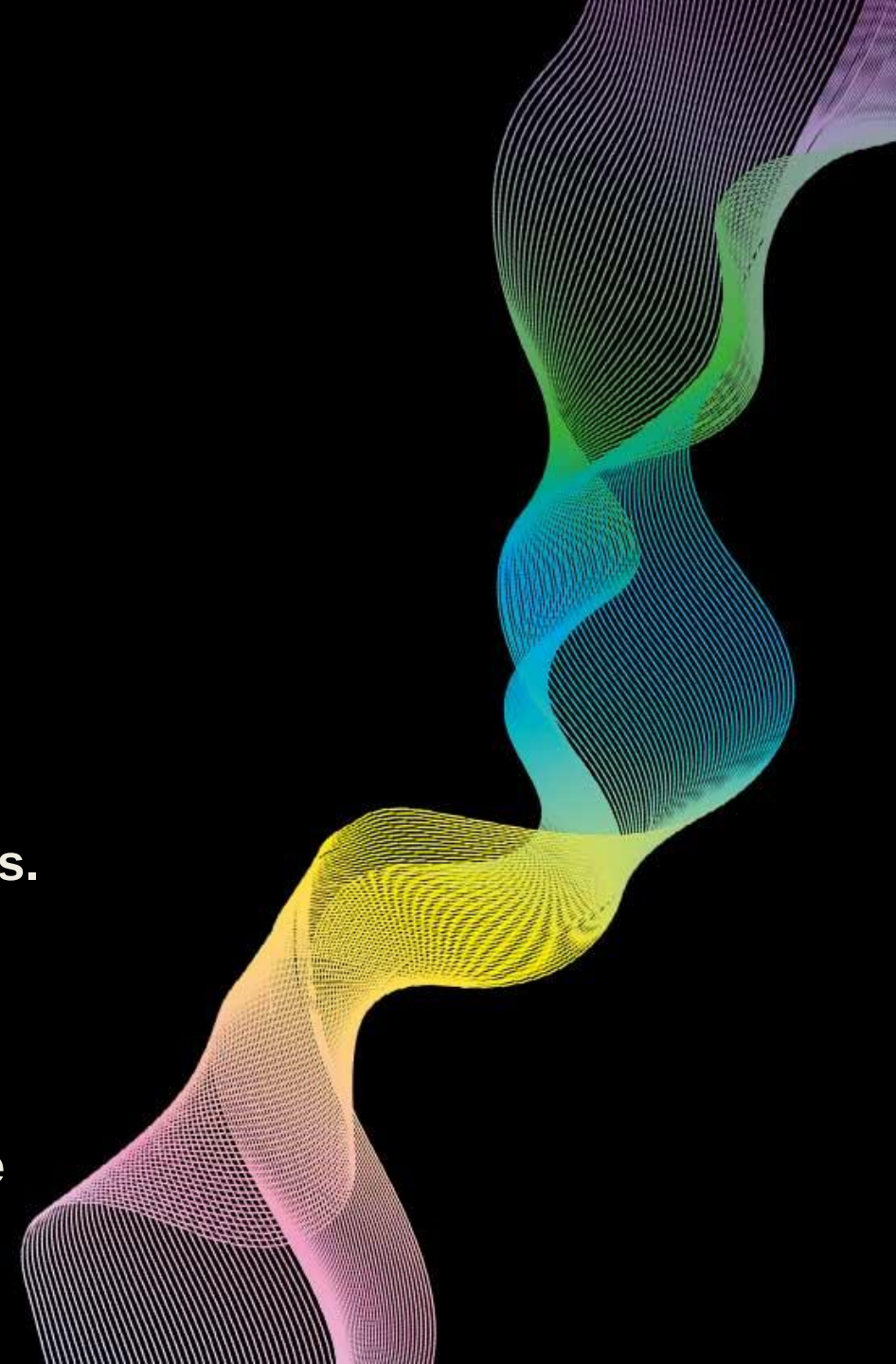
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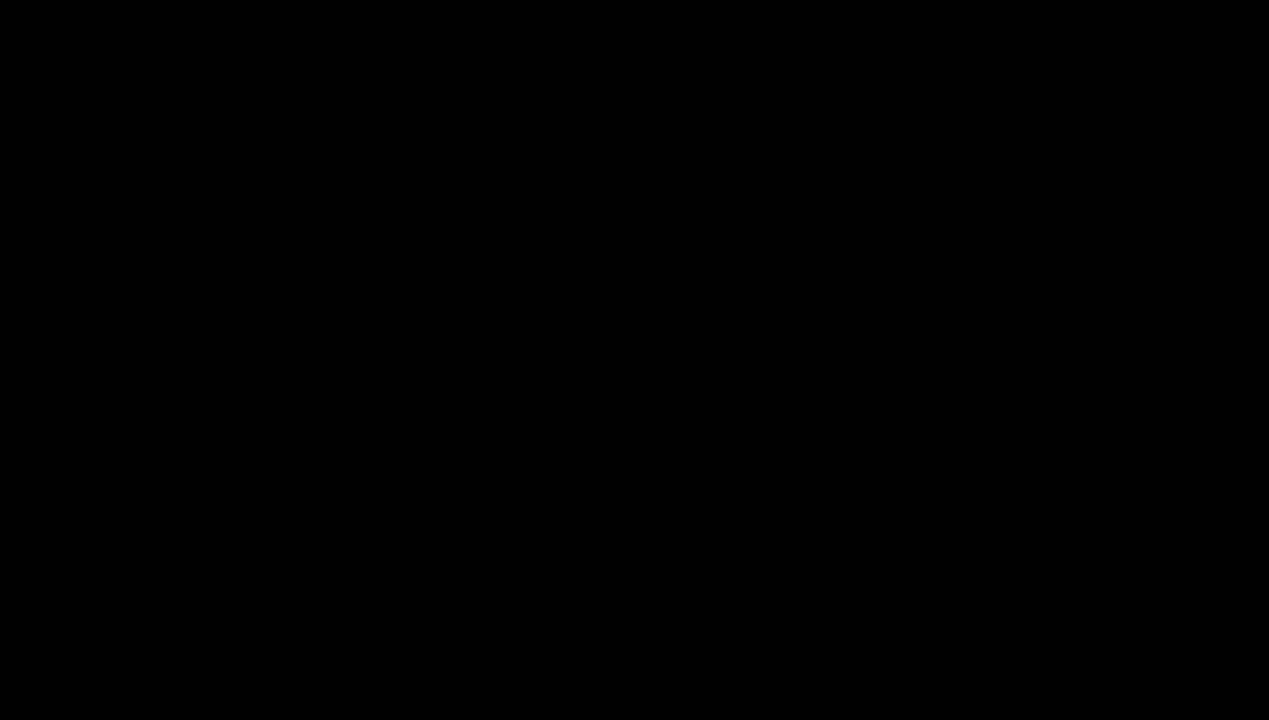
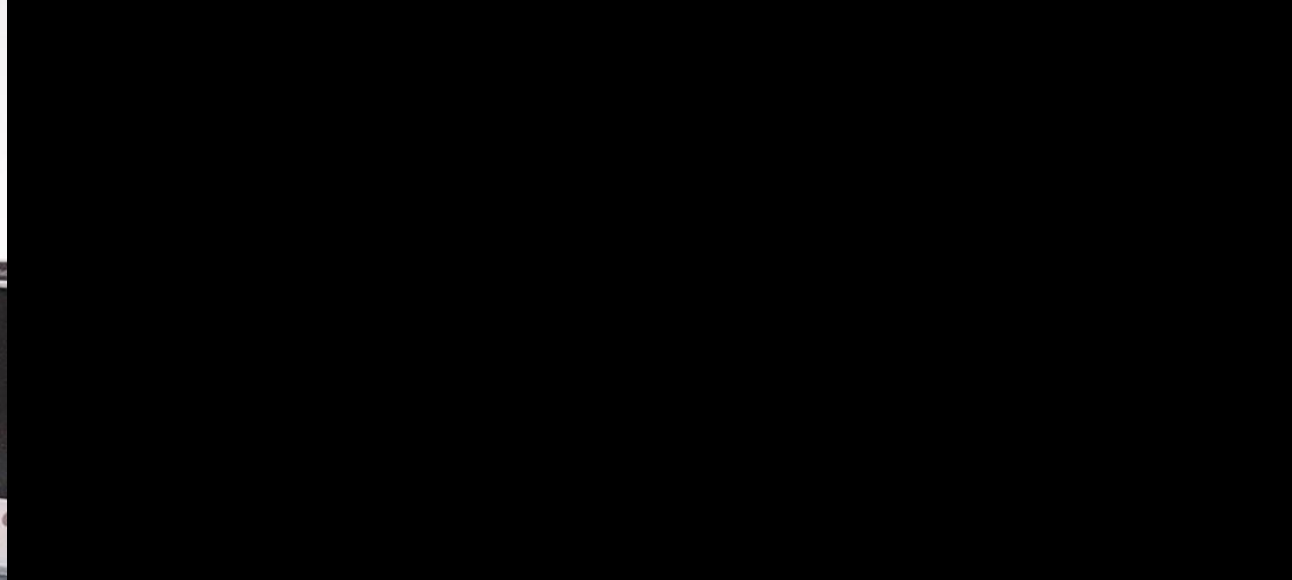
Examples of Human-Centered Design Successes

Healthcare Devices with User-Friendly Interfaces

- **Blood Glucose Monitors:** Devices used by individuals with diabetes to monitor blood glucose levels have evolved to become more user-friendly. Many now have clear displays, simple navigation, and reminders for testing, making them accessible to a wide range of users.
- **Medical Imaging Equipment:** Modern medical imaging devices, such as MRI and CT scanners, have improved user interfaces that help technicians operate them efficiently. Intuitive controls and user guidance enhance the overall experience for healthcare professionals and patients.



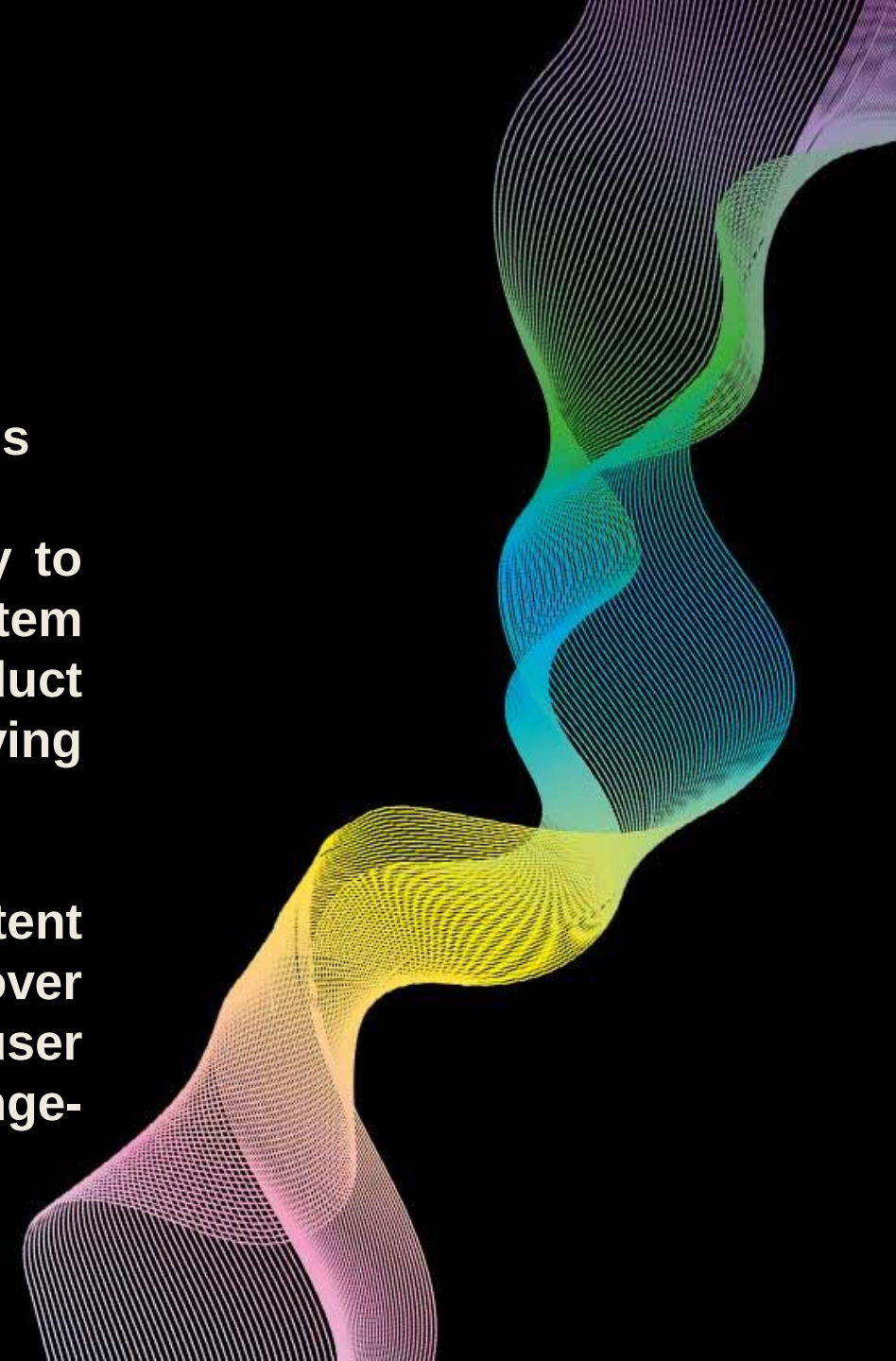




Examples of Human-Centered Design Successes

E-commerce Platforms with personalized recommendations

- **Amazon:** Amazon employs HCD principles extensively to enhance the user experience. Its recommendation system uses machine learning to provide personalized product suggestions, improving customer satisfaction and driving sales.
- **Netflix:** Netflix uses HCD to personalize content recommendations, making it easier for users to discover movies and TV shows aligned with their interests. The user interface is designed for ease of navigation and binge-watching.



amazon

Deliver to
Los Angeles 90011

All



Hello, Sign in
Account & Lists

Returns
& Orders



All

Best Sellers

Amazon Basics

New Releases

Today's Deals

Customer Service

Prime

Books

Registry

Gift Cards

Toys & Games

Fashion

Kindle Books

Amazon Home

Watch *The Wheel of Time*



Watch characters
as unique as you are

prime video



Cozy winter style for her



Coats and jackets



Accessories



Boots



Sweaters

Bring in the fresh energy



Fitness equipment



Yoga equipment



Health & personal care



Outdoors

Must-see fashion deals



All deals



Clothing



Handbags



Shoes

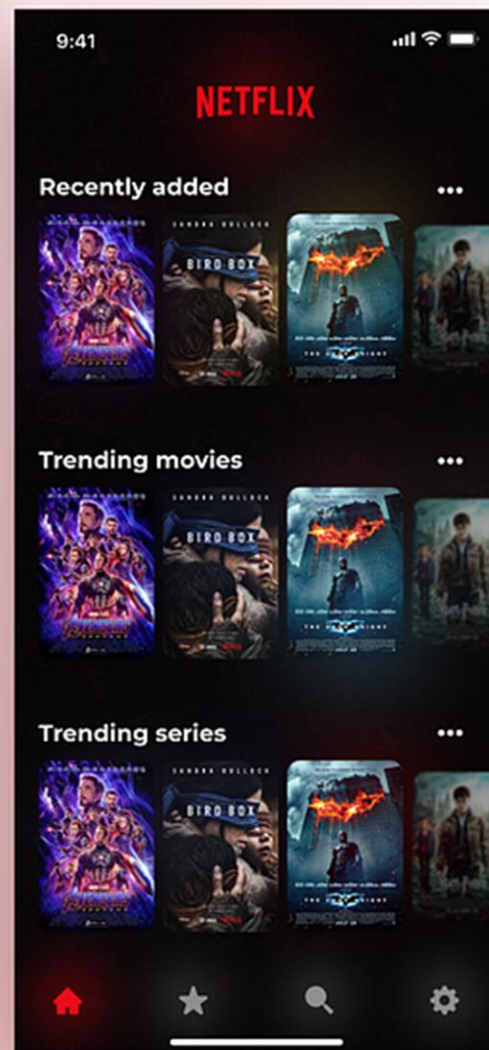
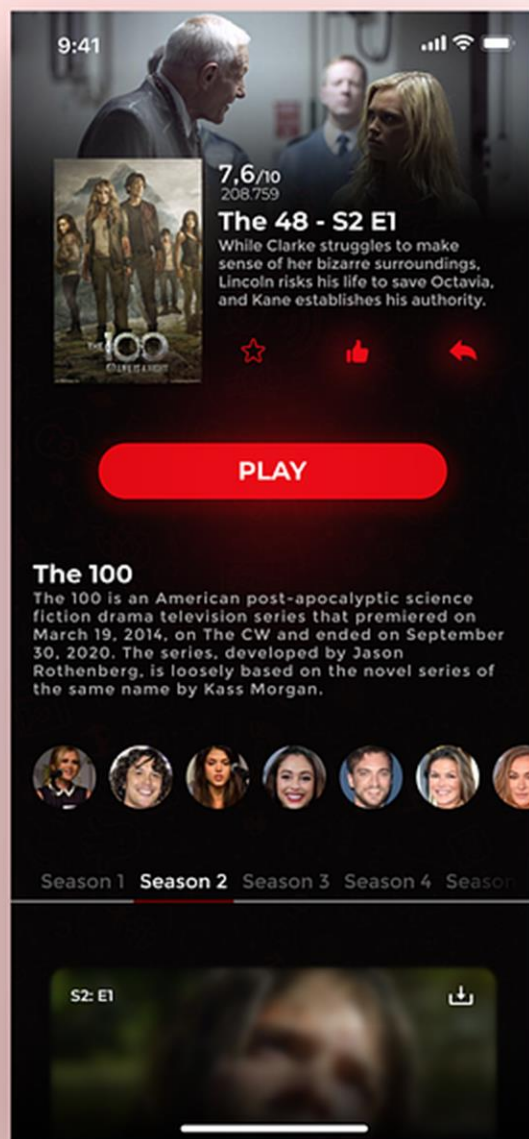
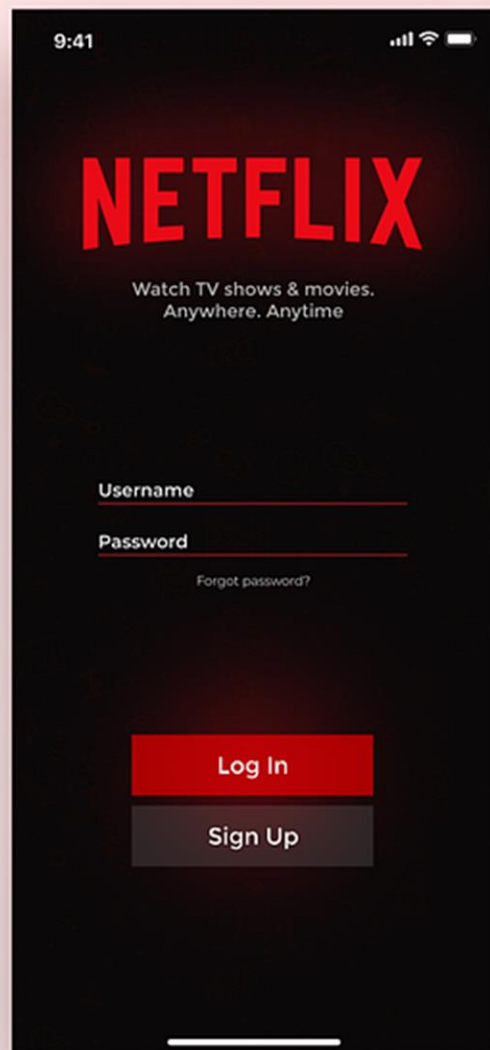
Sign in for the best experience

Sign in securely

Grant a few
birthday wishes

Shop gift cards

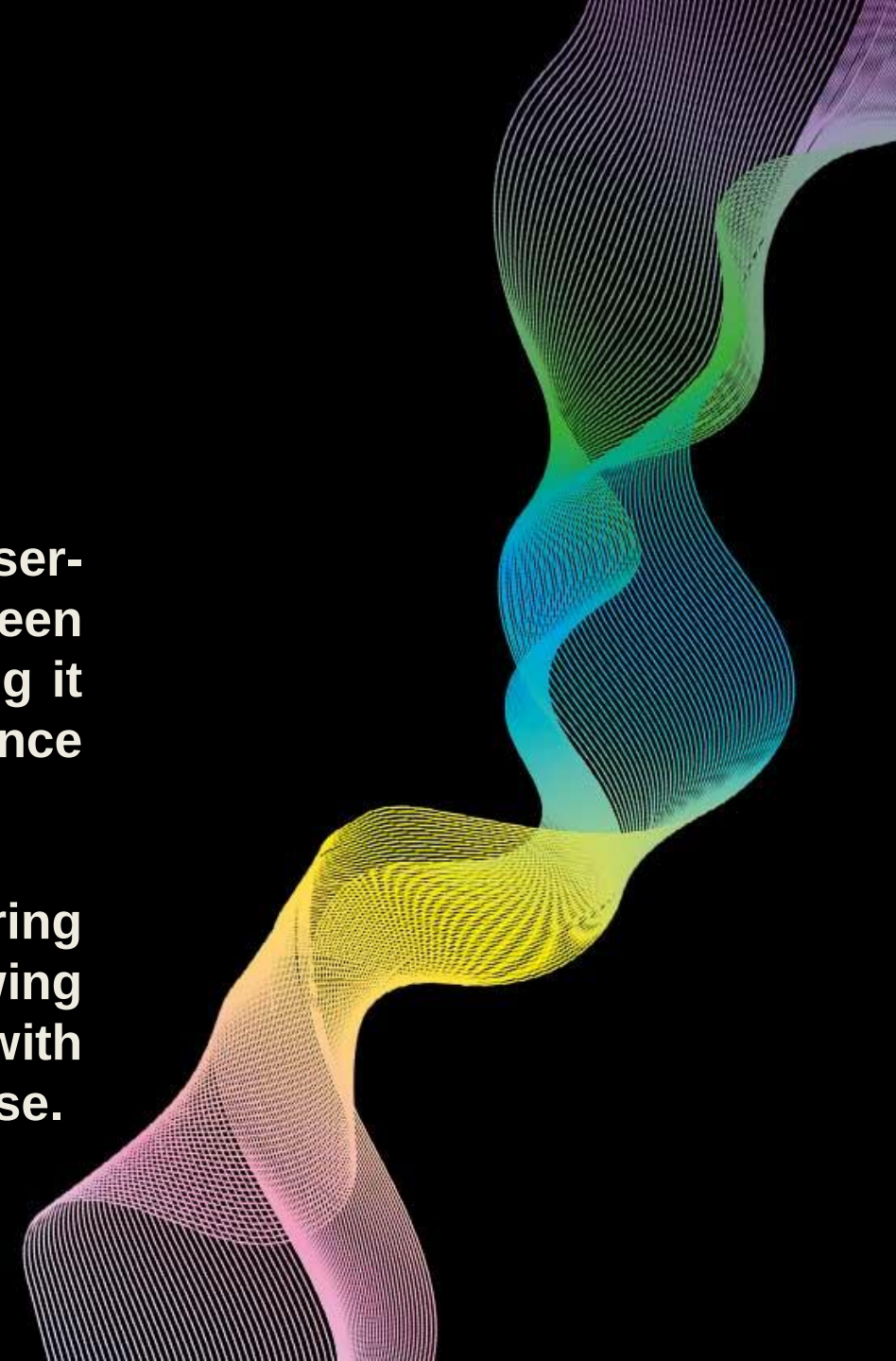




Examples of Human-Centered Design Successes

Automobile Interiors and Infotainment Systems

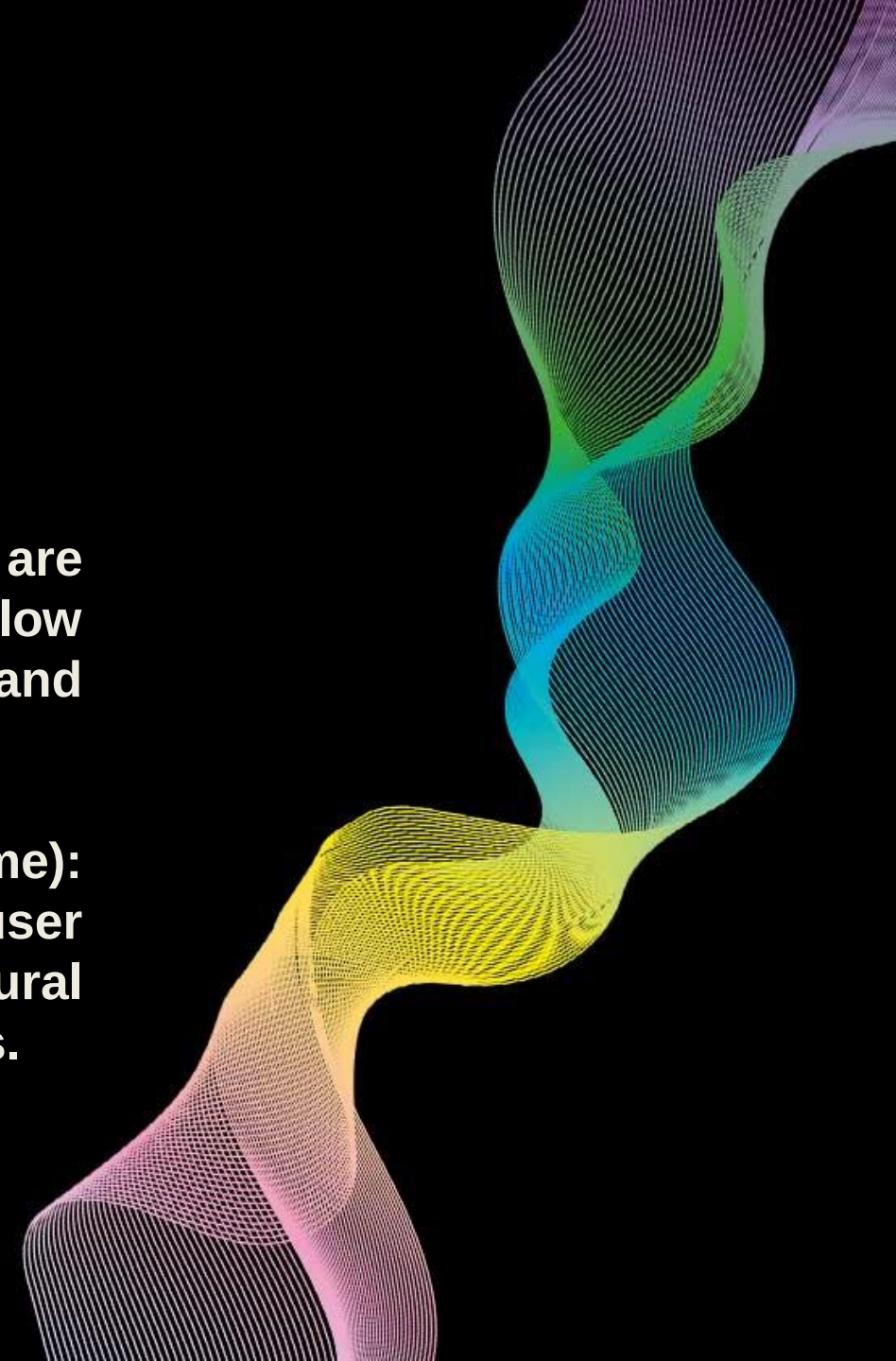
- **Tesla:** Tesla's electric vehicles are known for their user-centric designs. The interior features a large touchscreen display that controls various aspects of the car, making it intuitive for drivers. Frequent software updates enhance the user experience over time.
- **Apple CarPlay and Android Auto:** These systems bring smartphone interfaces to the car's dashboard, allowing drivers to access maps, music, and other apps with minimal distraction. They prioritize safety and ease of use.



Examples of Human-Centered Design Successes

Smart Home Devices

- **Smart Thermostats (e.g., Nest, Ecobee):** These devices are designed with user-friendly interfaces that allow homeowners to easily control and schedule heating and cooling, leading to energy savings and comfort.
- **Voice Assistants (e.g., Amazon Echo, Google Home):** Voice-controlled smart speakers prioritize user convenience. They are designed to understand natural language and provide helpful responses to user queries.

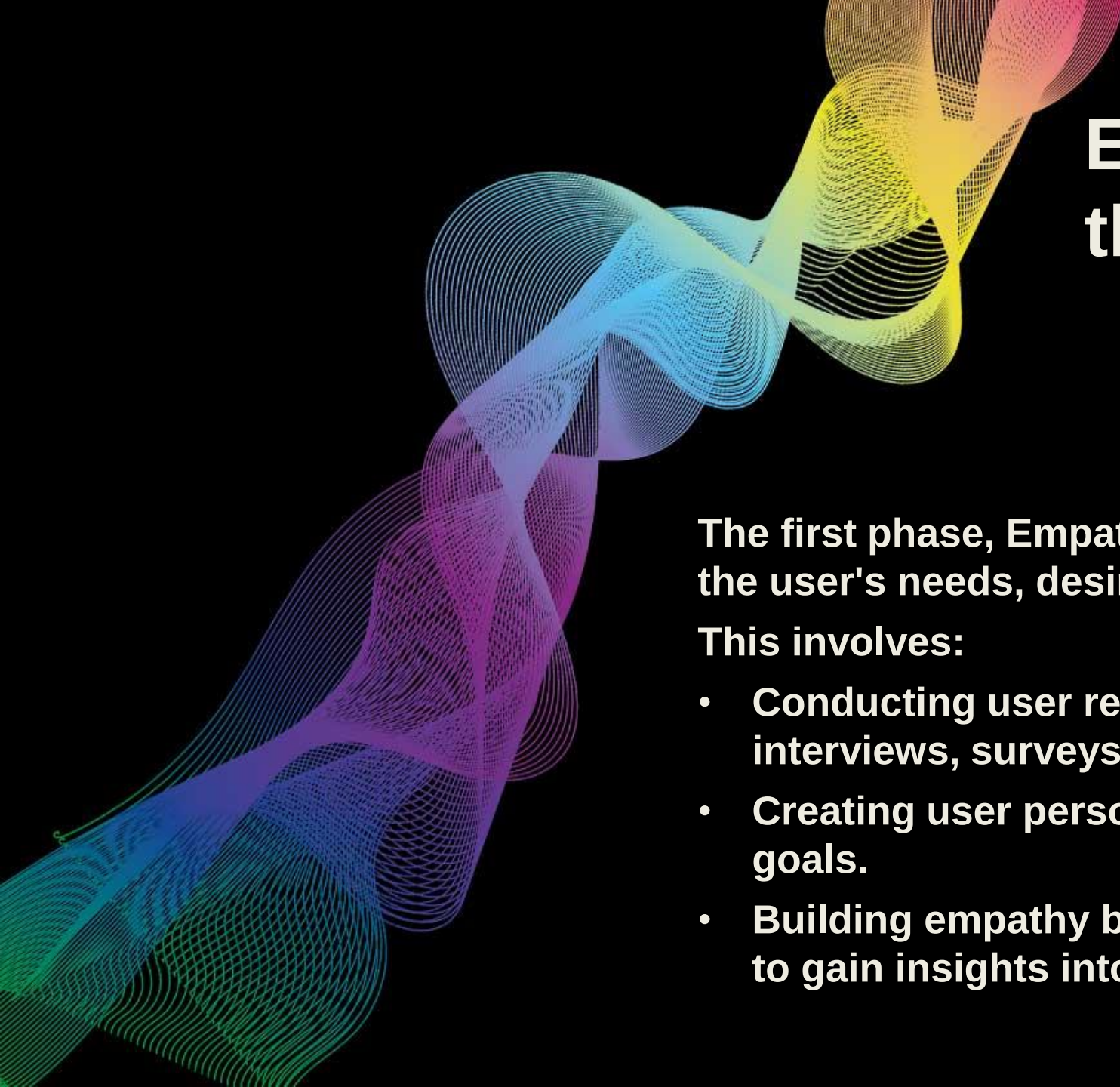






Three Main Phases of Human-Centered Design



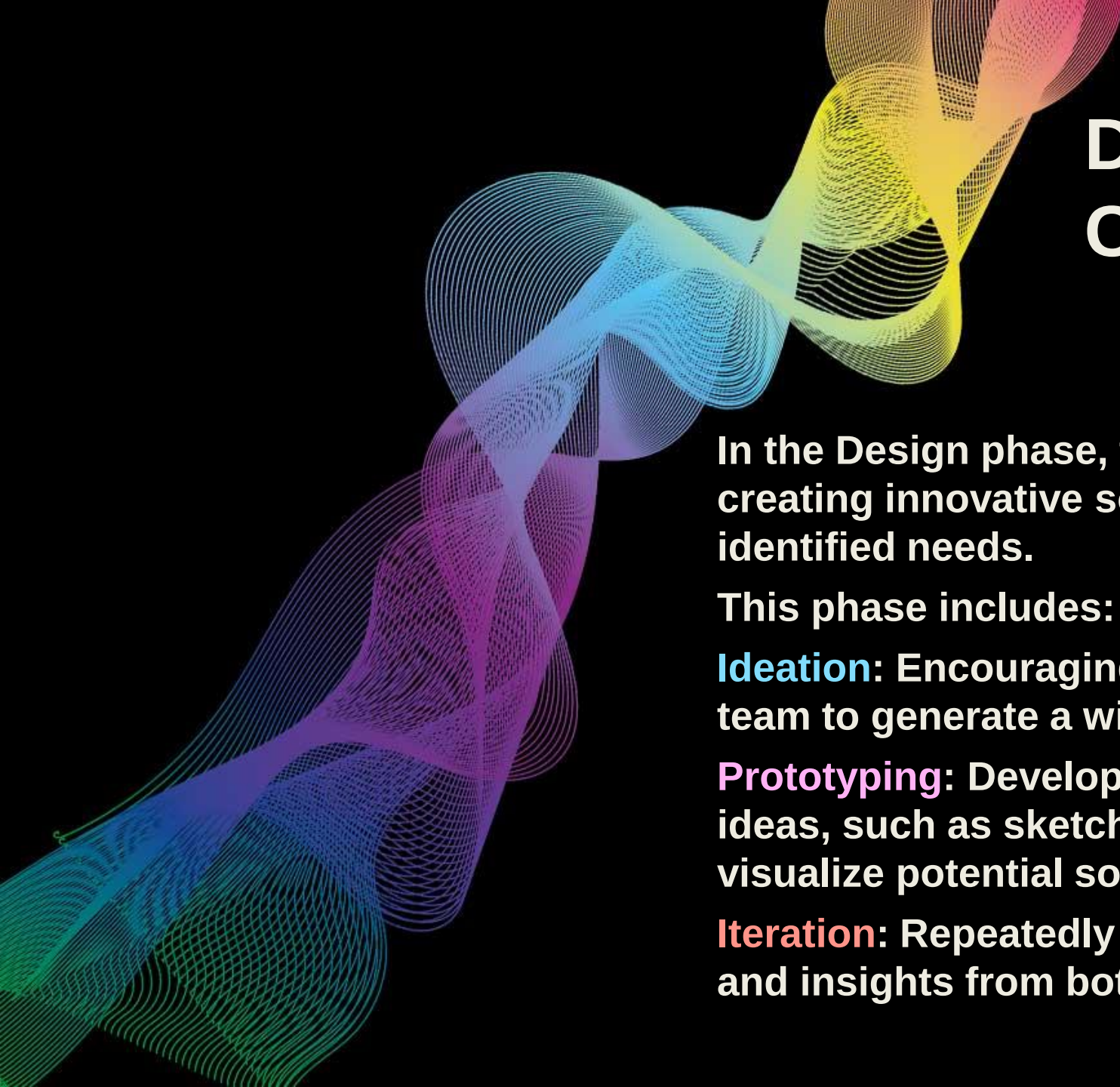


Empathize: Understand the User's Perspective

The first phase, Empathize, is about deeply understanding the user's needs, desires, and challenges.

This involves:

- **Conducting user research through methods like interviews, surveys, observations, and user testing.**
- **Creating user personas to represent typical users and their goals.**
- **Building empathy by putting yourself in the user's shoes to gain insights into their experiences.**



Design: Generate Creative Solutions

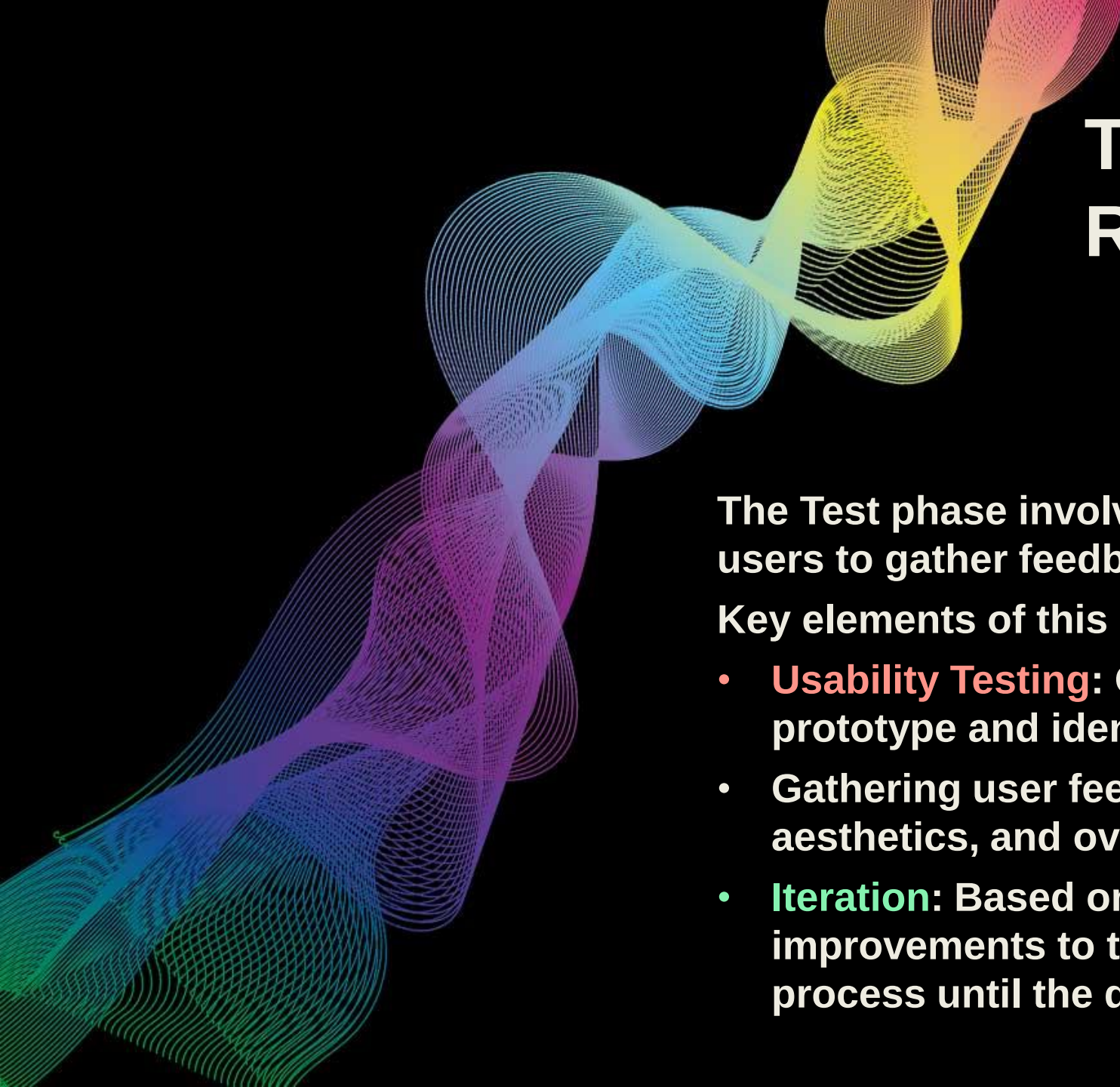
In the Design phase, the focus shifts to brainstorming and creating innovative solutions that address the user's identified needs.

This phase includes:

Ideation: Encouraging creative thinking within the design team to generate a wide range of ideas.

Prototyping: Developing low-fidelity representations of these ideas, such as sketches, wireframes, or mock-ups, to visualize potential solutions.

Iteration: Repeatedly refining the design based on feedback and insights from both the team and potential users.



Test: Validate and Refine the Design

The Test phase involves putting the prototypes in front of real users to gather feedback and insights.

Key elements of this phase include:

- **Usability Testing:** Observing how users interact with the prototype and identifying areas of difficulty or confusion.
- Gathering user feedback on the design's functionality, aesthetics, and overall experience.
- **Iteration:** Based on user feedback, making necessary improvements to the design, retesting, and repeating the process until the design meets user expectations.

Empathize phase

The Empathize phase is the foundational step in Human-Centered Design (HCD) and sets the stage for a user-centric approach to product or system development.

During this phase, the primary objective is to gain a deep and comprehensive understanding of the end-users' needs, desires, and behaviors.



Empathize phase

This phase is characterized by:

User Research: In the Empathize phase, designers and researchers engage in various user research activities, such as interviews, surveys, focus groups, and observations.

User Personas: A user persona is a fictional representation of a typical user, and it includes demographic information, motivations, goals, pain points, and behaviors. Personas help in humanizing and visualizing the user base, making it easier for the design team to empathize with their needs.

Empathy Building: Empathy is the cornerstone of the Empathize phase. Designers aim to develop a profound sense of empathy for the users, effectively "walking in their shoes" to understand their experiences and emotions. This emotional connection is crucial for creating solutions that genuinely address user problems.

Problem Statement: This statement should be based on the insights gained from user research and should define the problem from the user's perspective.



Empathize phase

Importance of Understanding User Needs and Behaviors: :

User-Centric Solutions: for designing solutions that are tailor-made for the intended users. It ensures that the end product or system that fulfills their goals.

Minimizing Assumptions: design based on concrete information. This reduces the risk of creating solutions that don't resonate with users.

Enhancing User Satisfaction: Meeting user needs leads to higher levels of user satisfaction

Reducing Revisions: can save time and resources in later phases. By gaining a deep understanding of user needs upfront, design revisions and changes are less likely to be required in later stages of the project.

Innovation and User Delight: Empathizing with users can lead to innovative solutions that not only fulfill their needs but also surprise and delight them.



Test phase

The Empathize phase is the foundational step in Human-Centered Design (HCD) and sets the stage for a user-centric approach to product or system development.

During this phase, the primary objective is to gain a deep and comprehensive understanding of the end-users' needs, desires, and behaviors.



Test phase

This phase is characterized by:

Purpose of the Test Phase: the critical stage where designers assess how well their design solutions align with user needs and expectations. It's the ultimate evaluation step to ensure the product or system's user-friendliness.

User-Centered Evaluation: The Test phase is user-centered, meaning it revolves around real users interacting with the product or system. The goal is to identify usability issues, uncover pain points, and gauge overall user satisfaction.

Usability Testing: Usability testing is a core component of the Test phase. It involves observing users as they complete tasks with the product, recording their interactions, and collecting feedback about their experiences.



Test phase

Gathering User Feedback and Iteration:

User Feedback: through surveys, questionnaires, interviews, and observation, designers capture user insights regarding their interactions with the product.

Identifying Pain Points: designers pay close attention to the pain points that users encounter. These pain points can range from difficulties in navigation, confusion about features, or any other obstacles to achieving their goals.

Data Analysis: This step often involves quantifying aspects like task success rates and time on task.

Iteration and Improvement: The Test phase is not a one-time activity but a catalyst for iteration. Designers use the gathered feedback to refine the design

Continuous Refinement: Subsequent iterations build on previous findings, leading to a design that becomes progressively more user-friendly

Iterative Nature of HCD

Emphasizing that HCD is an Iterative Process:

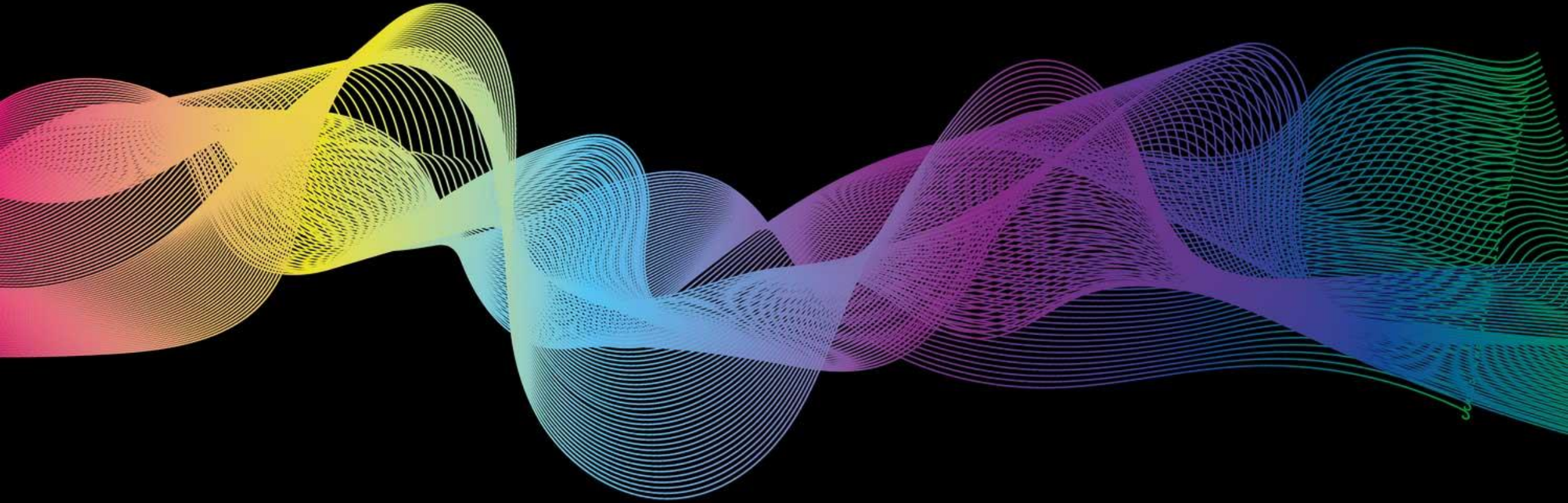
- **Revisiting Phases:** HCD is not a linear process but a cycle that often involves revisiting the phases. After completing the initial cycle of Empathize, Design, and Test, it's common to start again with new insights or adjustments.
- **Iterative Refinement:** Each iteration serves to refine and enhance the design. In subsequent cycles, designers incorporate feedback and lessons learned, resulting in an improved and user-focused solution.
- **Feedback Loop:** HCD is marked by a continuous feedback loop with the end-users. This feedback loop helps designers make necessary changes, validate assumptions, and ensure that the product or system remains aligned with user needs.

Iterative Nature of HCD

Continuous Improvement through Feedback

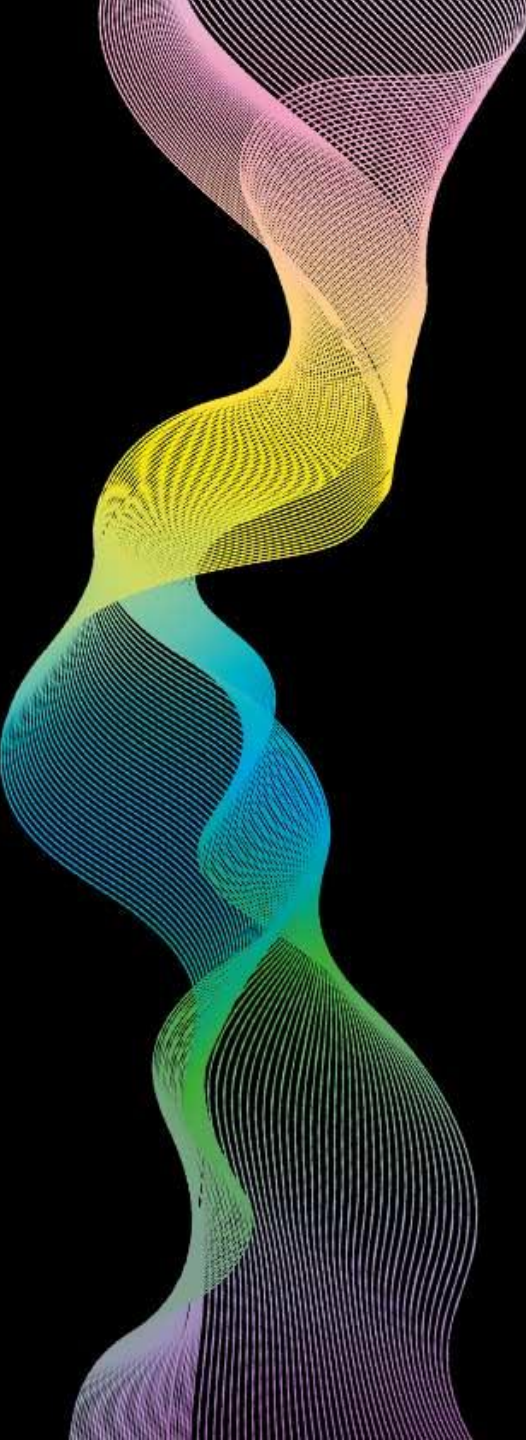
- **User-Centered Decision Making** : *ensures that design choices are validated by those who will use the product*
- **Adapting to Changing Needs** : *allows the design to adapt to these changes, ensuring that the product remains relevant*
- **Minimizing Risk** : *designers can identify and address issues early, reducing the risk of major design flaws*
- **Enhanced Quality** : *provides insights that lead to refinements*
- **Optimizing Resources** : *It prevents large-scale rework by catching issues early*
- **Innovation and Creativity** : *It fosters an environment of continuous learning and adaptation, which can lead to breakthrough solutions*
- **Cultural Change** : *foster a culture of continuous improvement*

USER-CENTERED DESIGN vs SYSTEM-DESIGNED SYSTEM

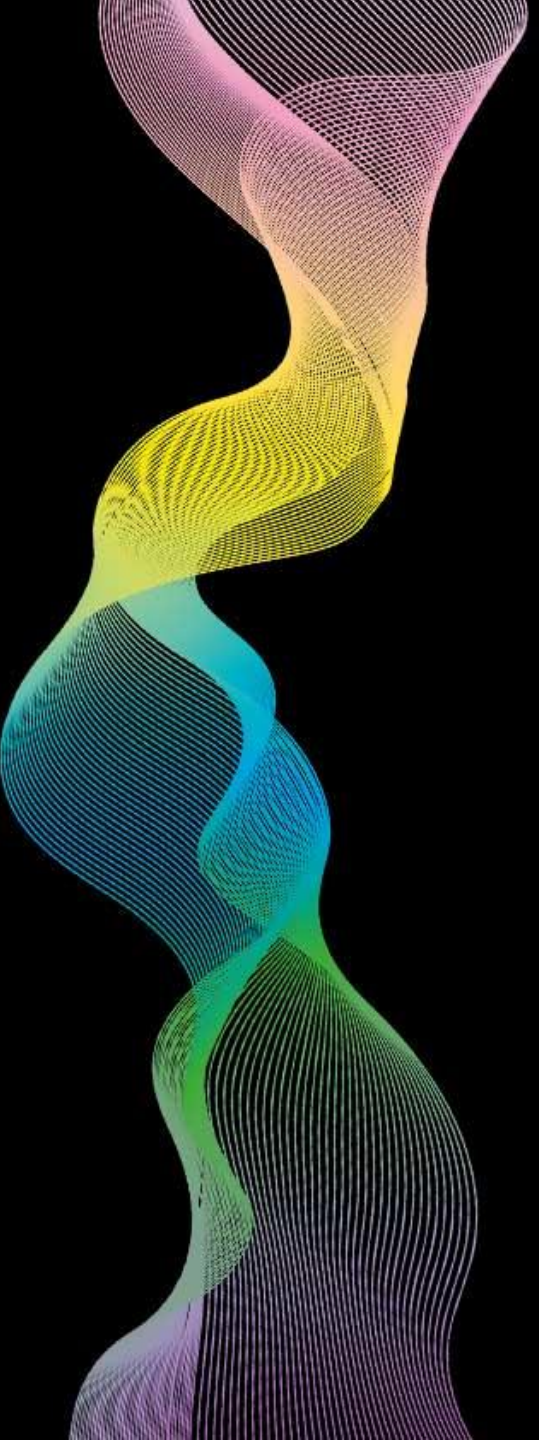


User-Centered Design (UCD) and System-Centered Design are two contrasting approaches to product and system development.



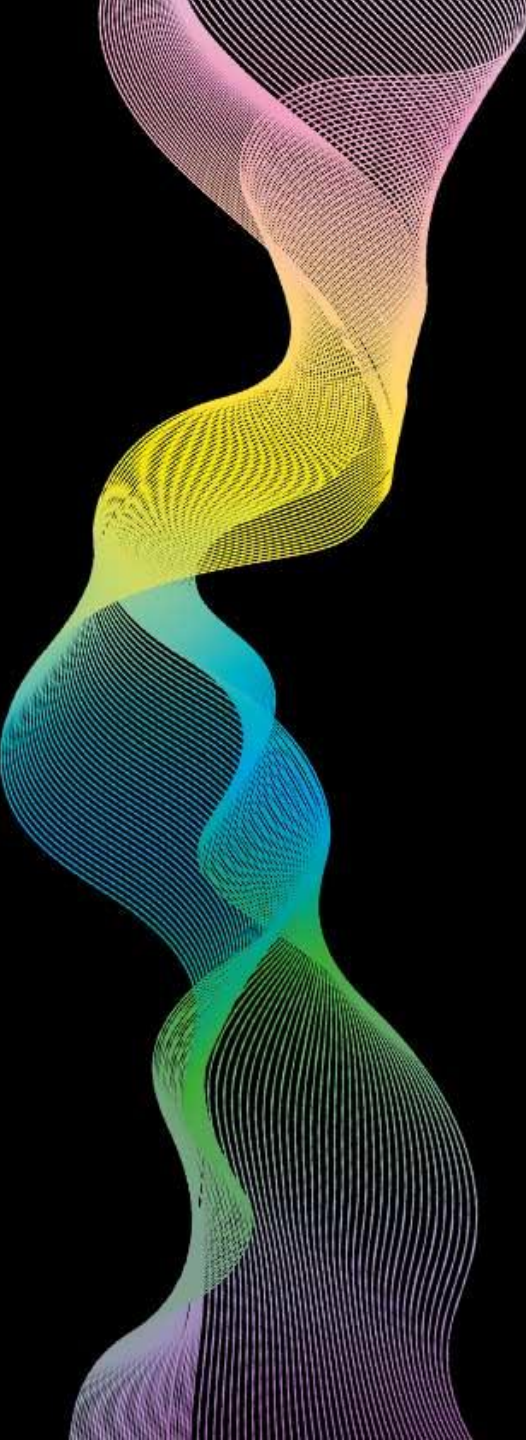


User-Centered Design (UCD) is an approach to product and system design that places the needs, preferences, and behaviors of the end-users at the forefront of the development process.



User Empathy: UCD begins with understanding the target user base. Designers conduct research, interviews, and observations to gain insights into user needs, desires, and pain points.

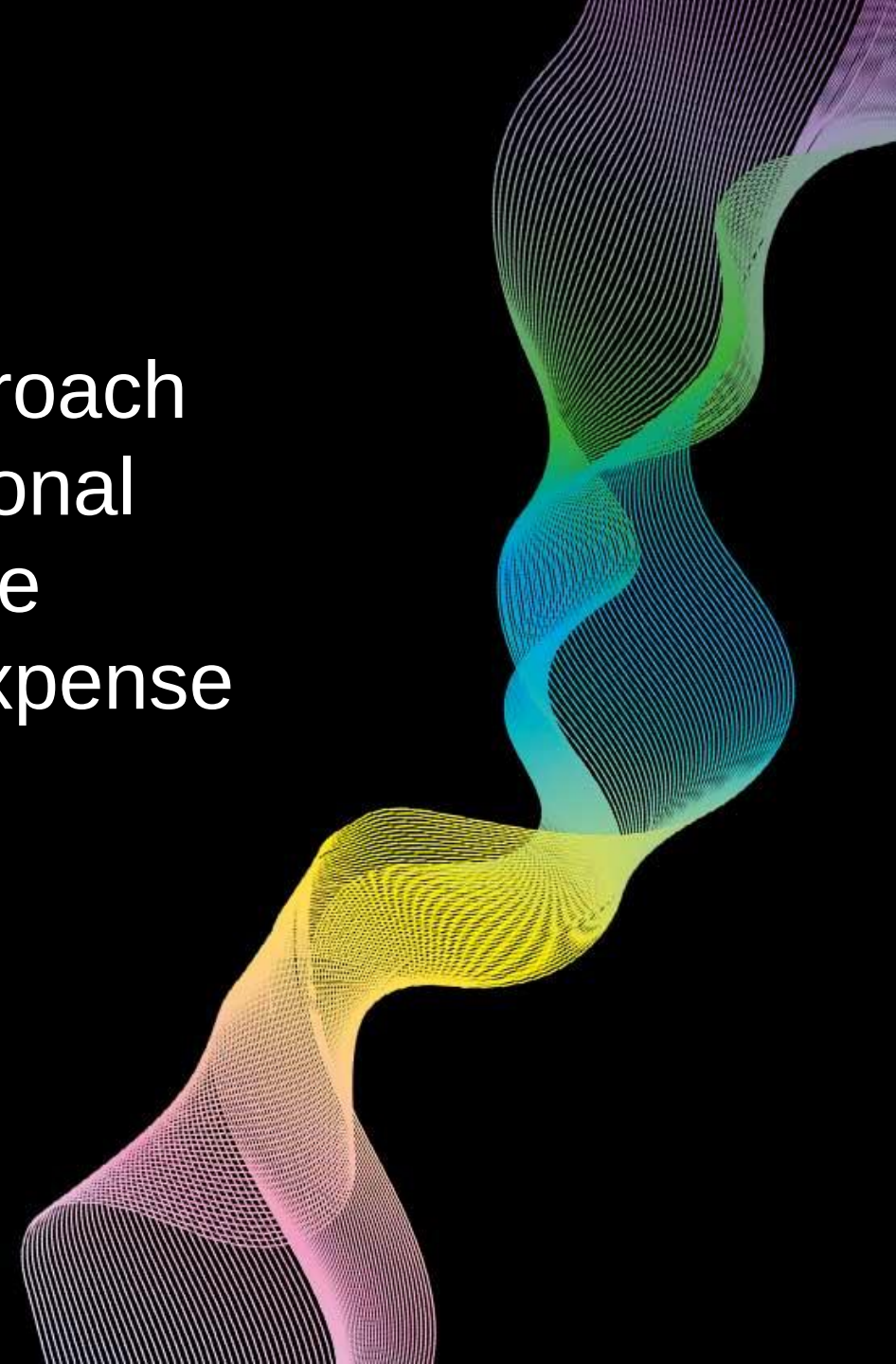
Iterative Process: UCD is an iterative process that involves continuous user feedback and testing. It aims to refine and improve the design through multiple iterations, ensuring that it evolves to meet users' evolving needs.



Usability and User Satisfaction: The primary focus of UCD is to create products that are usable and provide a positive user experience. It prioritizes elements like intuitive interfaces, clear navigation, and user satisfaction.

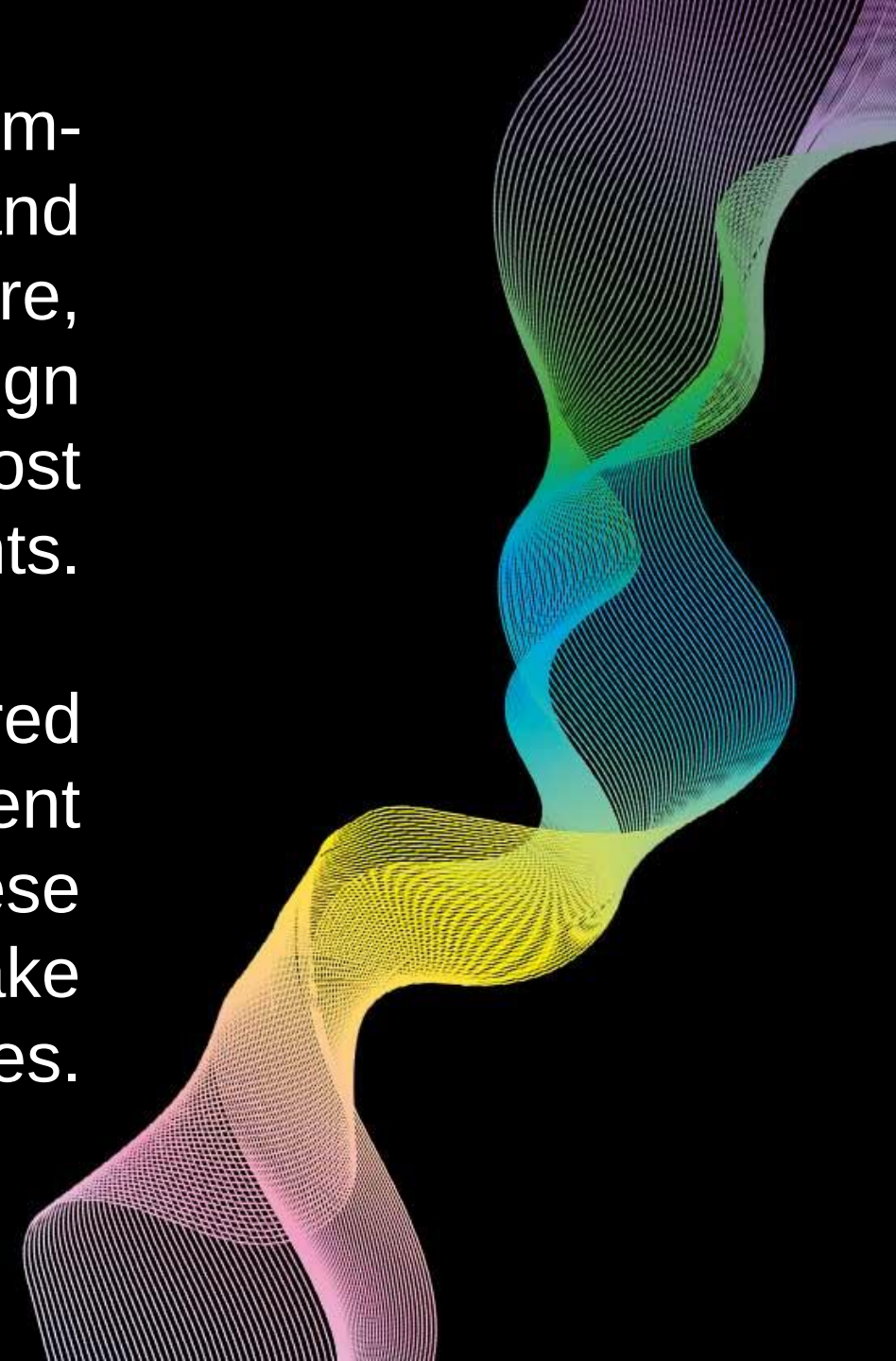
User-Centric Decision Making: Design choices are made based on user feedback, rather than assumptions. The ultimate goal is to create a product that aligns with the users' goals and enhances their lives.

System-Centered Design is an approach that places technical or organizational constraints at the forefront of the development process, often at the expense of user needs.



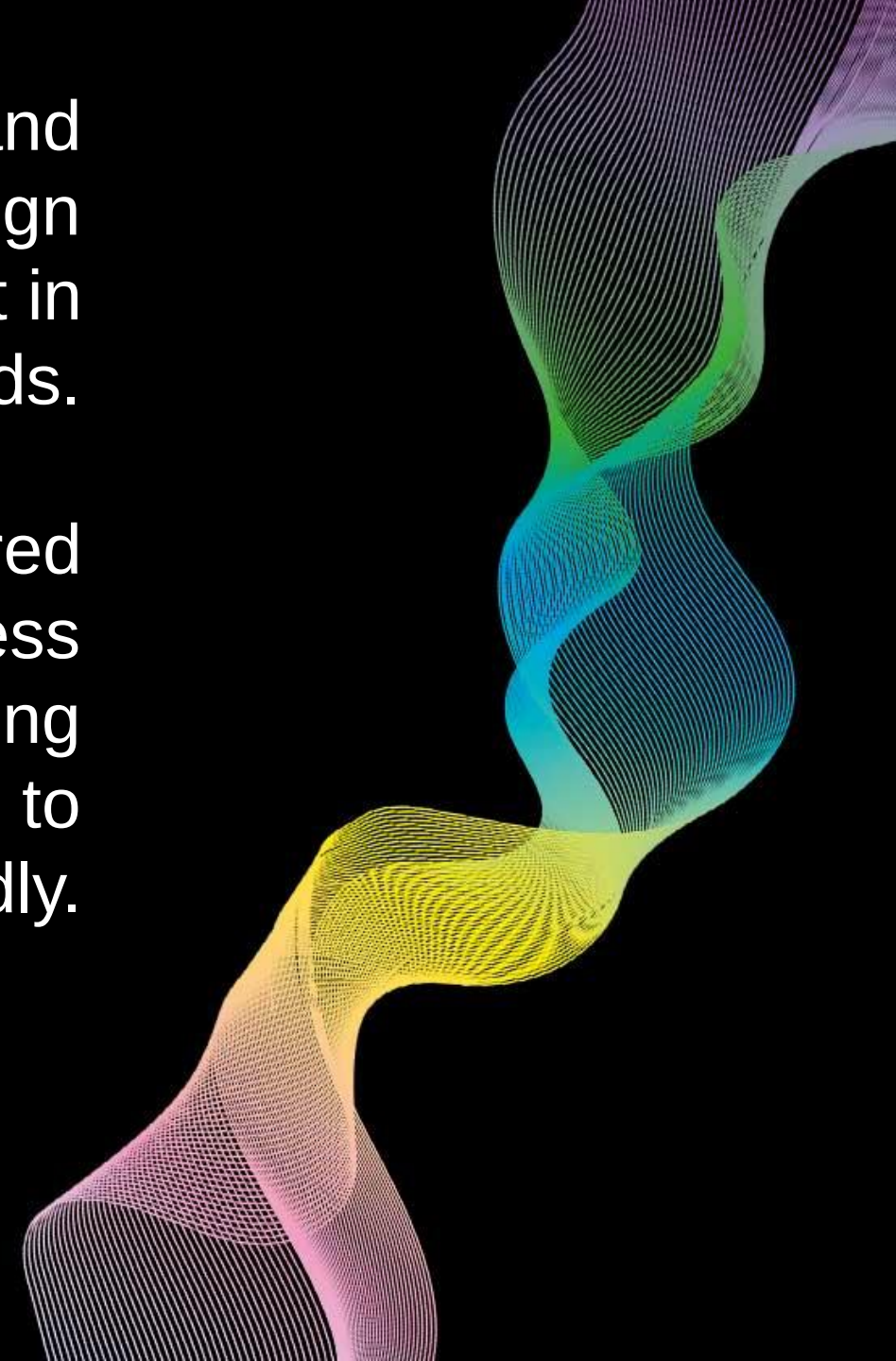
Technical and Organizational Focus: System-Centered Design prioritizes the constraints and requirements of the technical infrastructure, hardware, or organizational goals. Design decisions are often made based on what's most feasible within these constraints.

Efficiency and Functionality: System-Centered Design aims to create solutions that are efficient and meet technical requirements. While these considerations are important, they sometimes take precedence over user-friendly interfaces.



Limited User Involvement: User feedback and testing are typically conducted later in the design process, if at all. This approach can result in products that are out of touch with user needs.

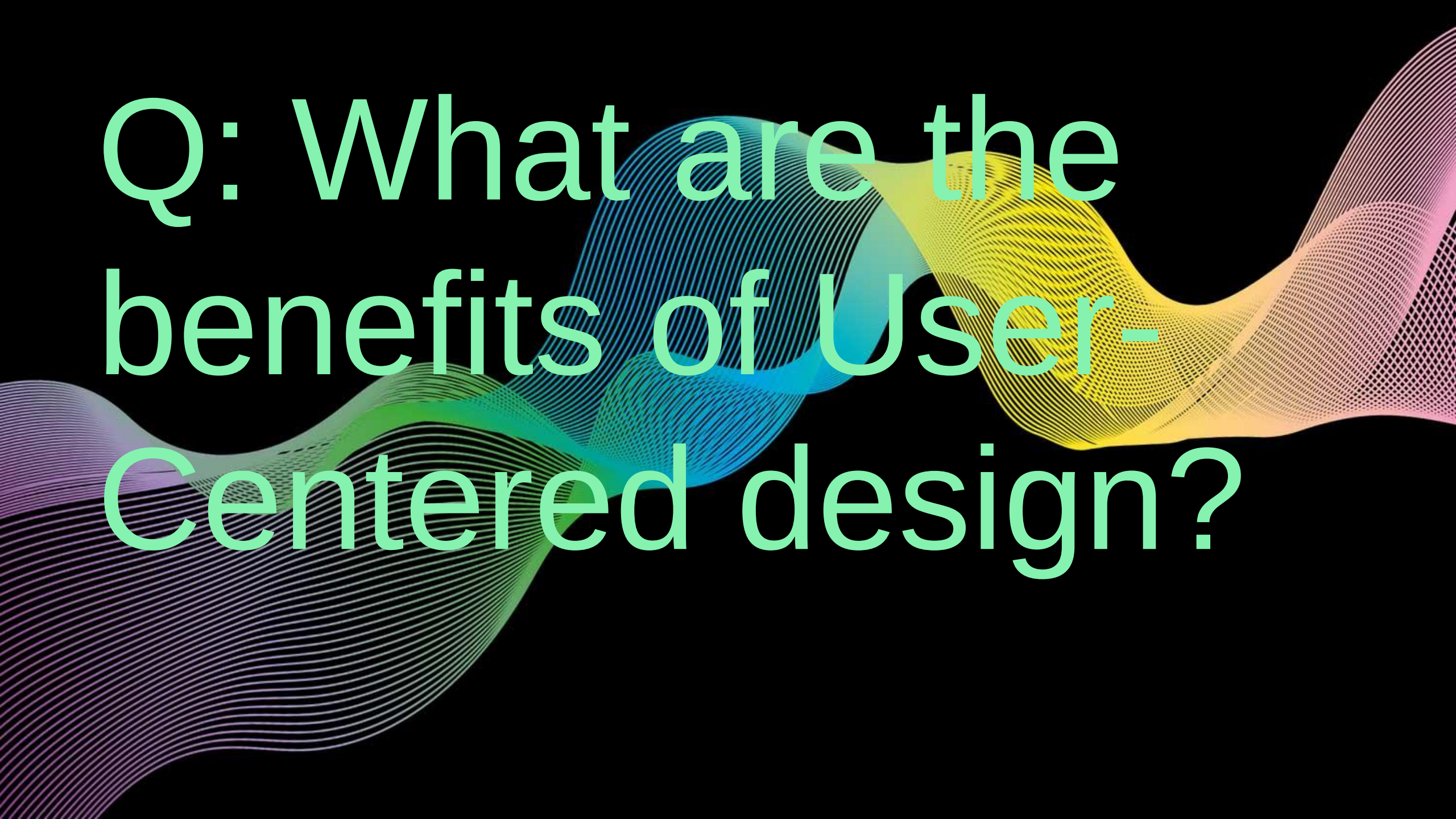
Rigid and Linear Process: System-Centered Design often follows a linear development process where design choices are made at the beginning and adhered to throughout, even if they prove to be less user-friendly.



UCD vs System centered design

- User-Centered Design prioritizes users' needs, iterative refinement, usability, and user satisfaction.
- System-Centered Design prioritizes technical and organizational constraints, efficiency, and functionality, sometimes at the expense of user-friendliness.



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Q: What are the
benefits of User-
Centered design?

Q: Can you think of a product or service that you've personally used or interacted with that you believe embodies User-Centered Design principles?

What specific aspects of its design make it user-friendly, and how have these design elements contributed to your positive experience with the product or service?

How have these features contributed to the product's success in terms of user adoption and satisfaction?

Product: WhatsApp

User-Friendly Aspects:

Simplicity: WhatsApp's design is simple and clean. It's easy to understand, with basic features like texting and calling front and center. You don't need a manual to use it.

Easy Navigation: The app's layout makes it clear where to find your chats, contacts, and settings. Icons are often universal, like the phone icon for calls.

Frequent Updates: WhatsApp regularly updates the app based on user feedback, adding features that people find useful.

Positive Experience:

Ease of Use: Because WhatsApp is straightforward, people from various age groups and backgrounds can use it without feeling overwhelmed.

Communication: It's an excellent tool for staying in touch with friends and family. You can chat, call, or video call, making it a one-stop communication platform.

User Feedback: WhatsApp listens to its users and often fixes issues or adds features based on their needs and requests.

User Adoption and Satisfaction:

Widespread Use: WhatsApp's user-friendly design and reliable communication features have led to widespread adoption. Many people worldwide use it daily.

High Satisfaction: Users like the app because it's easy to use, works well, and has become a part of their daily lives. The positive user experience contributes to its high user satisfaction and loyalty.

Product: Mobile banking app

User-Friendly Aspects:

Simplified Access: Mobile banking apps often offer biometric login options like fingerprint or face recognition, making it easy and secure to access your account.

Intuitive Interface: The app's interface is designed to be user-friendly, with easily recognizable icons for checking balances, transferring funds, and paying bills.

Quick Actions: Many mobile banking apps provide quick actions like depositing checks by taking a photo, which saves time and reduces the need to visit a physical bank.

Positive Experience:

Convenience: Users can manage their finances on the go, making banking more convenient and flexible.

Security: The app typically includes security features like alerts for unusual account activity.

Savings: By enabling users to set up automatic savings or budget tracking, the app helps people manage their money more effectively.

User Adoption and Satisfaction:

High Adoption: Mobile banking apps are widely adopted because they offer a convenient and secure way to handle financial transactions.

User Trust: Banks invest in user-friendly features and security to build trust and encourage users to rely on the app for their banking needs.

High Satisfaction: Users appreciate the convenience and features of the app, leading to high satisfaction and loyalty.

Product: Spectacles

User-Friendly Aspects:

Comfortable Fit: Spectacles are designed to fit comfortably on your face. They come in different sizes and shapes to suit various people.

Clear Vision: They help people see better. If you need prescription glasses, they are made to your specific needs, so you can see clearly.

Style Choices: Spectacles come in various styles, colors, and materials to match personal preferences.

Positive Experience:

Improved Vision: When you wear spectacles that fit and correct your vision, it's easier to read, work, or enjoy the view.

Fashion and Style: You can choose spectacles that match your style, making you feel confident and comfortable wearing them.

Customization: Prescription spectacles are personalized for your eyes, so you get the best vision possible.

User Adoption and Satisfaction:

Widespread Use: Many people wear spectacles every day to help them see better, so they're widely adopted.

High Satisfaction: People who wear spectacles are often very satisfied because they improve their daily lives, and the variety of styles allows them to express their personality.

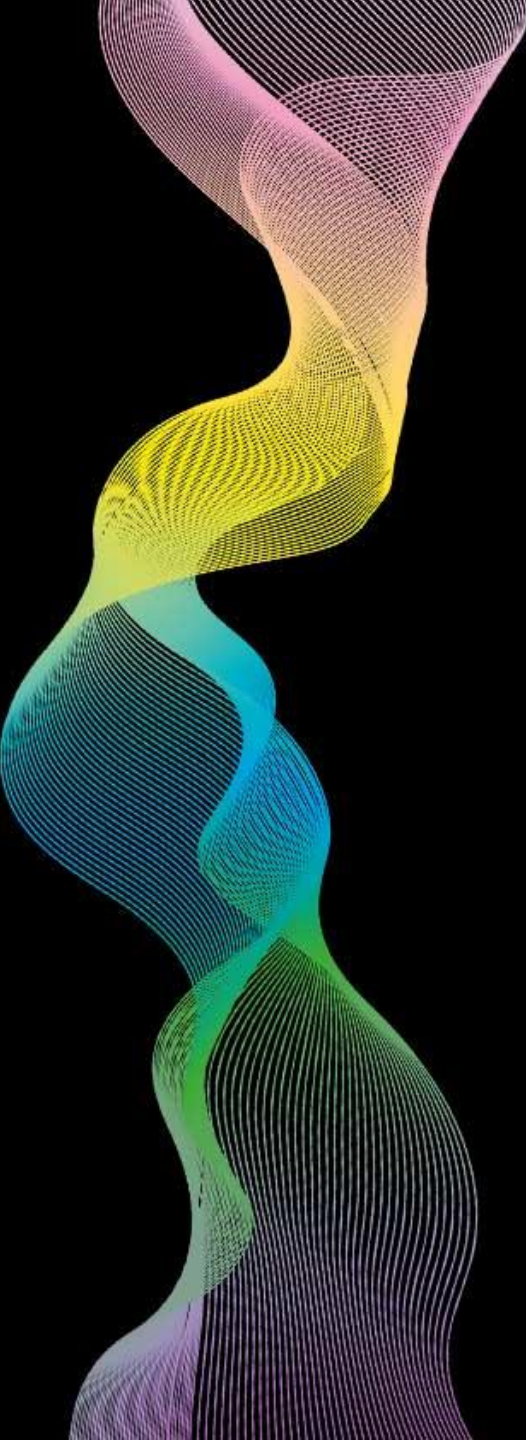
Q: Why do you think some products or services prioritize system constraints over user-friendliness? Can you provide an example of a situation where a focus on system constraints led to challenges or user dissatisfaction?

Technical limitations, cost, company rules and regulations (industrial), legacy systems(old computers in use)

There are many examples related to security that can be addressed, for e.g recent elevator security in skyscrapers need access card/pin to use elevator.

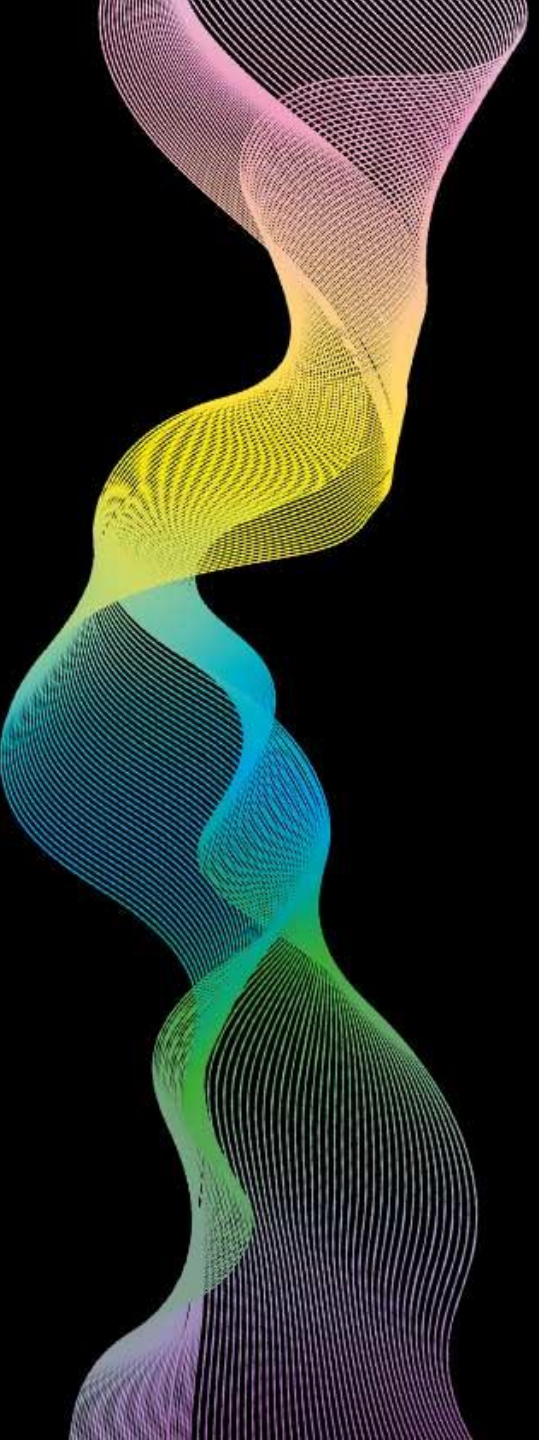
Automatic car washing: follow certain sequence of steps to wash

Automated customer service – through phone.



User Experience (UX)

User Experience (UX) refers to the overall emotional, cognitive, and practical response a user has while interacting with a product, system, or service. It encompasses how a user perceives, feels, and interacts with a design, and it is shaped by various factors, including usability, aesthetics, accessibility, and more.



User Experience consists of several key elements, each contributing to the overall quality of the experience:

Usability: The product's ease of use and efficiency.

Accessibility: The product's ability to be used by people with various abilities.

Aesthetics: The visual appeal and design of the product.

Utility: How well the product serves the user's needs and goals.

Credibility: The trustworthiness and believability of the product or service.

Satisfaction: The user's overall contentment and delight with the experience.

Emotion: The emotional impact of the interaction.

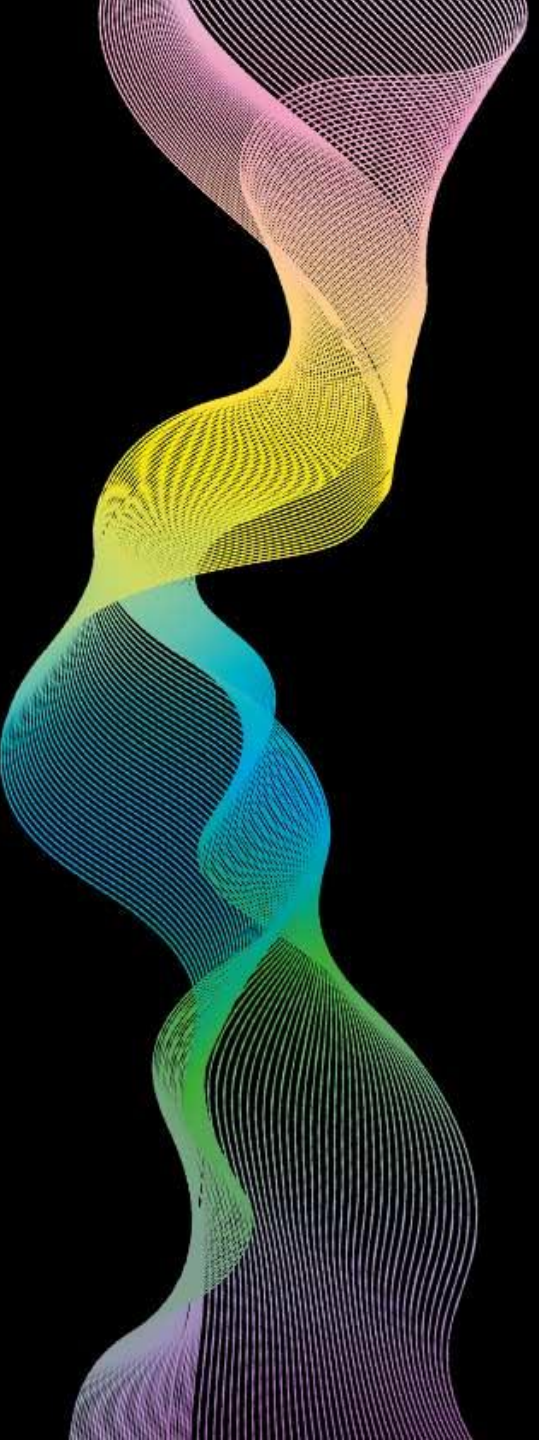
Learnability: How easy it is for users to understand and begin using the product.

Efficiency: How quickly and accurately users can complete tasks.

Memorability: How easily users can remember how to use the product.

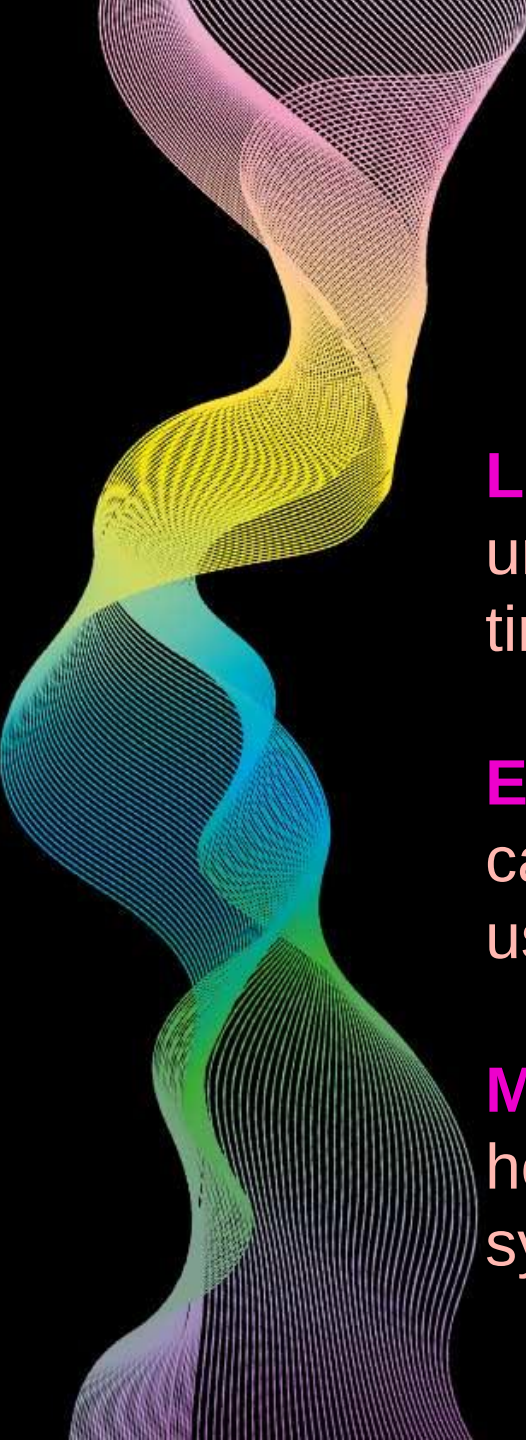
Errors: The frequency and severity of user errors during interaction.

Engagement: The extent to which the user is captivated by the experience.

An abstract graphic on the left side of the slide, consisting of a wavy, multi-colored line that flows vertically. The line is composed of many thin, overlapping lines, creating a sense of movement and depth. The colors transition from purple at the top, through yellow and green, to blue and teal at the bottom.

Usability

Usability is a key component of User Experience (UX). It focuses on the ease with which a user can learn, interact with, and navigate a product or system to achieve their goals. Usability assesses how effectively and efficiently users can complete tasks and how satisfied they are with the experience.

A decorative graphic on the left side of the slide consisting of several overlapping, wavy, vertical lines. The lines are colored in a gradient from purple at the top, through yellow and green, to blue at the bottom. They have a fine, mesh-like texture.

Usability principles

Learnability: It refers to how easily users, including novices, can understand and begin using a product. Learnability minimizes the time and effort required for users to become proficient.

Efficiency: Efficiency assesses how quickly and accurately users can accomplish tasks within the product. Efficient systems save users time and reduce frustration.

Memorability: Memorability gauges how well users can remember how to use a product after an initial interaction. A highly memorable system reduces the need for re-learning.



Usability principles

Errors: Evaluating the number, severity, and types of user errors is critical. A usable design minimizes the occurrence and impact of errors, leading to a smoother user experience.

Satisfaction: Usability also considers user satisfaction. A user who enjoys interacting with a product is more likely to use it regularly and recommend it to others. Satisfaction is often linked to user loyalty.



Measuring Usability

Usability Testing: Usability testing involves observing real users as they interact with a product or system.

It provides firsthand insights into how users navigate and experience the design.

Test participants are given tasks to complete, and their interactions are observed and analyzed.

Usability testing helps identify usability issues and informs design improvements.



Measuring Usability

Heuristic Evaluation: Heuristic evaluation involves usability experts (evaluators) assessing a product against a set of established usability principles or heuristics.

The evaluators identify potential issues based on their expertise. While it doesn't replace user testing, heuristic evaluation can quickly uncover usability problems in early design stages.

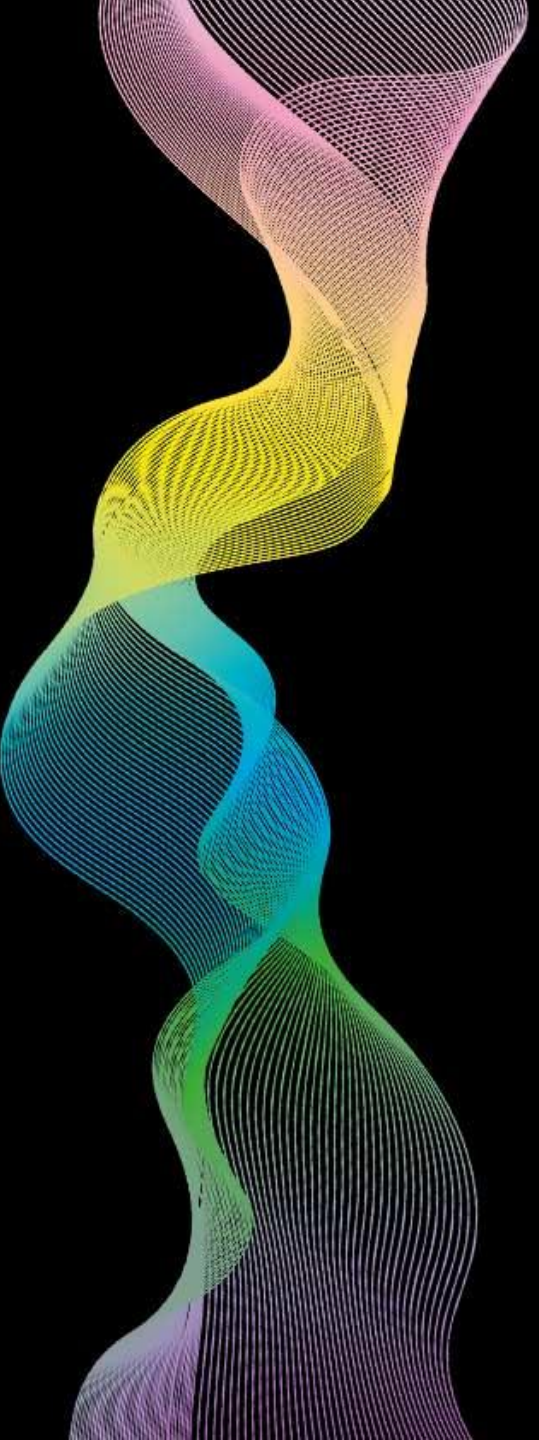


Measuring Usability

Surveys: Surveys are used to gather user feedback and perceptions about a product's usability.

They can be designed to measure user satisfaction, identify pain points, and assess the overall user experience.

Surveys can provide quantitative data and help gauge user sentiment.



An e-commerce website wants to assess the usability of its new checkout process. They invite a group of potential customers to complete a purchase on the website while their interactions are observed. Test participants are asked to add items to their cart, proceed to checkout, and make a simulated purchase.

How this scenario could be adapted for all three usability testing methods?